

Lecture Series 2: Statistical tests CM2018

Adam S. Darwich

darwich@kth.se

L2-2 On the relationships between variables


$\rho = corr(x, y)$
$cov(x, y)$
$Y = f(X_i, \beta) + e_i$

Correlation

$$\boldsymbol{\rho = corr(x, y)}$$

Covariance

$$\boldsymbol{cov(x, y)}$$

Regression

$$\boldsymbol{Y = f(X_i, \beta) + e_i}$$

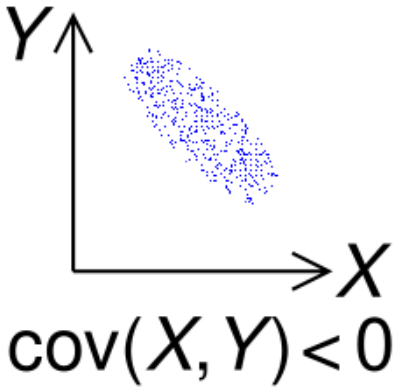
Covariance

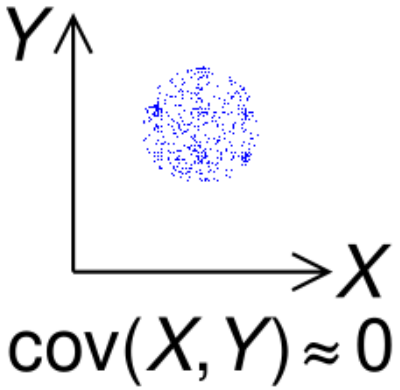
- A scale dependent measure of joint probability between two variables, *X* and *Y*.
- A measure of linear relationship where *Cov(X,Y)* < 0 indicates a negative relation, *Cov(X,Y)* = 0 indicates no relation, *Cov(X,Y)* > 0 indicates a positive relation.
- Variance is a special case of covariance where *X* = *Y*.

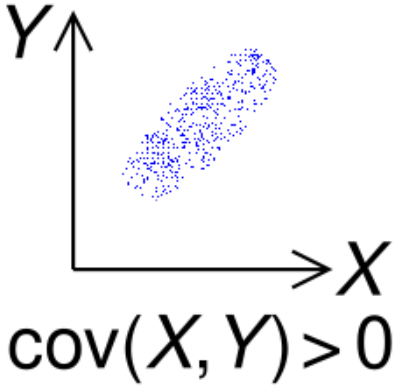
X	Y
6.1	7.1
6.2	6.4
1.5	2.0
…	…

$$Cov(X,Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$$

$$Cov(X,Y) = 7.35$$

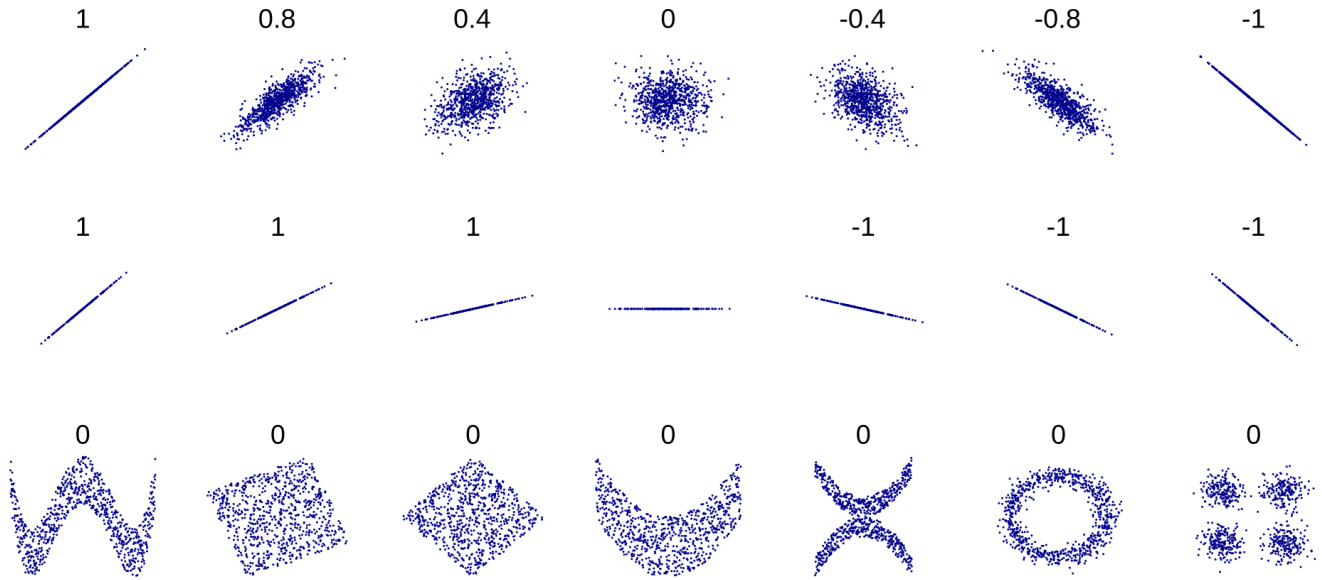






Correlation

- Paired, bivariate, data.
- Depenence between two random variables.
- Correlation is the standardised version of covariance.
- Degree of correlation measured using:
  - Pearson correlation coefficient (linear monotonic relationships),
  - Spearman's rank correlation coefficient, Kendall's tau (nonlinear monotonic relationships).
- The correlation is denoted ρ *or* r $(-1 \text{ to } 1)$.



X	*Y*
6.1	*7.1*
6.2	*6.4*
1.5	*2.0*
…	*…*

$$r_{Pearson,xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$$

Correlation

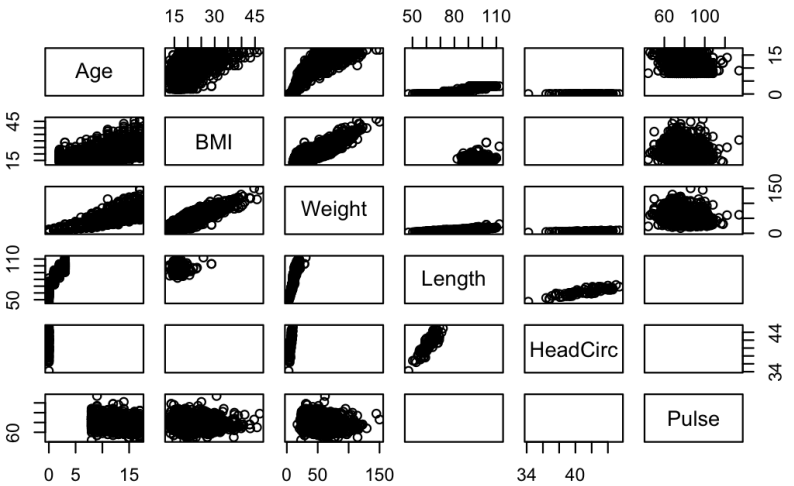
```r
## Binomial distribution
# we can use the binomial pmf to generate data for the success of a
# hypothetical surgery e.g., p: 0.981, n: 421 (based on Fazlinovic et al. 2021)
surgery_stats <- rbinom(421,1,0.981)
n_count <- length(surgery_stats)

# we can plot the data using for example a bar plot grouped by successes and failures:
count_outcome <- table(surgery_stats)
barplot(count_outcome, names=c("deceased","survived"), main = paste("30-day outcome
(N=",n_count,"patients)"), xlab = "Outcome", ylab = "Frequency")

# we can then get the proportions of the two outcomes
probs <- prop.table(count_outcome)

# the probability of survival is then:
p_success <- as.numeric(probs[2])

# we can then plot the probability mass function for a binomial distribution
# in this case, the probability of number of successes given 100 trials
pmf_binom <- dbinom(x = c(0:100), size = 100, prob = p_success)
plot(0:100, pmf_binom, type = "l", ylab ="Probabilities")
```



Variance-Covariance Matrices

```
cov(data,…)
> nhanes_cov
```

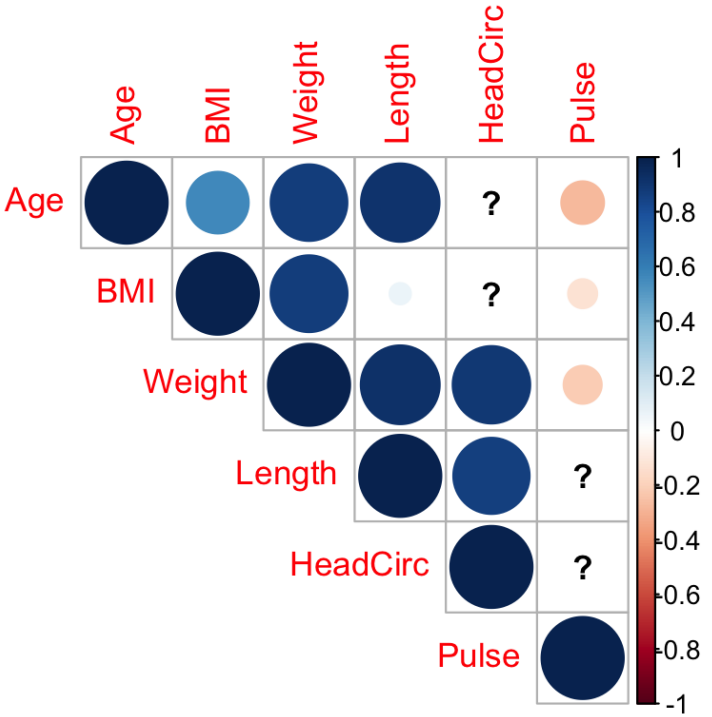
covariance matrix

	Age	BMI	Weight	Length	HeadCirc	Pulse
Age	27.536745	13.8735531	111.085203	13.8864468	0.000000	-9.041331
BMI	13.873553	27.6342269	107.969388	0.8341356	NA	-8.218937
Weight	111.085203	107.9693885	592.107689	48.1059045	2.978405	-51.007533
Length	13.886447	0.8341356	48.105905	187.8277317	9.894079	NA
HeadCirc	0.000000	NA	2.978405	9.8940792	5.342956	NA
Pulse	-9.041331	-8.2189373	-51.007533	NA	NA	140.243200

```
cor(data,…)
> nhanes_cor
```

correlation matrix

	Age	BMI	Weight	Length	HeadCirc	Pulse
Age	1.0000000	0.56574488	0.8712999	0.91291276	NA	-0.2702955
BMI	0.5657449	1.00000000	0.8759988	0.06664105	NA	-0.1249484
Weight	0.8712999	0.87599881	1.0000000	0.92196344	0.8902976	-0.2143250
Length	0.9129128	0.06664105	0.9219634	1.00000000	0.8619637	NA
HeadCirc	NA	NA	0.8902976	0.86196371	1.0000000	NA
Pulse	-0.2702955	-0.12494840	-0.2143250	NA	NA	1.0000000

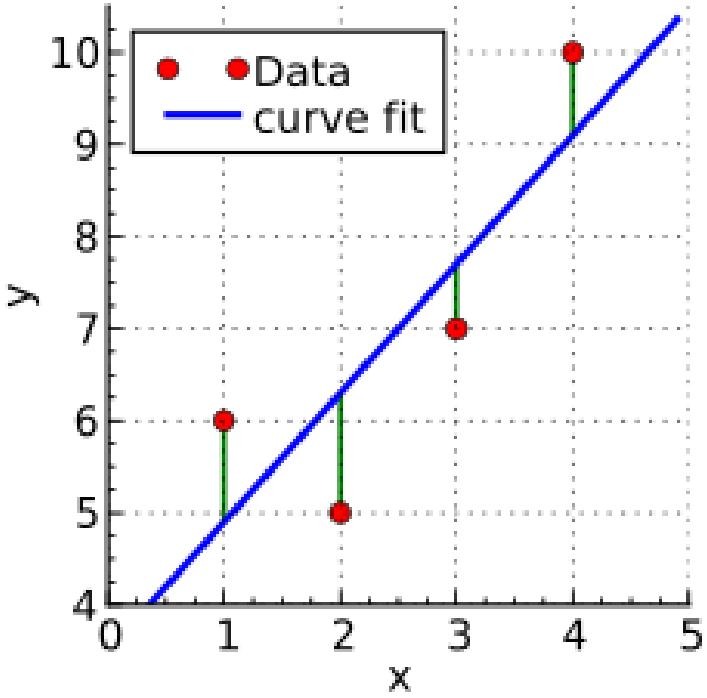



```
library(corrplot)
corrplot(data,…)
```

Linear regression more on this later

$$y = \beta_0 + X\beta + \varepsilon$$

- A model of the relationship between an dependent variable (*DV* or *Y*) and an independent variable (*IV* or *X*).
- The parameter β_0 or α describes *Y* at the intercept of *X=0*.
- The β coefficients (β_1 to β_n) indicate to what degree X_1 to X_n describe *Y* in a linear relationship.
- Used to make inferences about a population from a sample.
- We usually assign *DV* to be the effect we are interested in studying, the outcome variable. The *IV*s are then the effectors or predictors.





L2-3 Linear Regression

Regression



ANTHROPOLOGICAL MISCELLANEA.

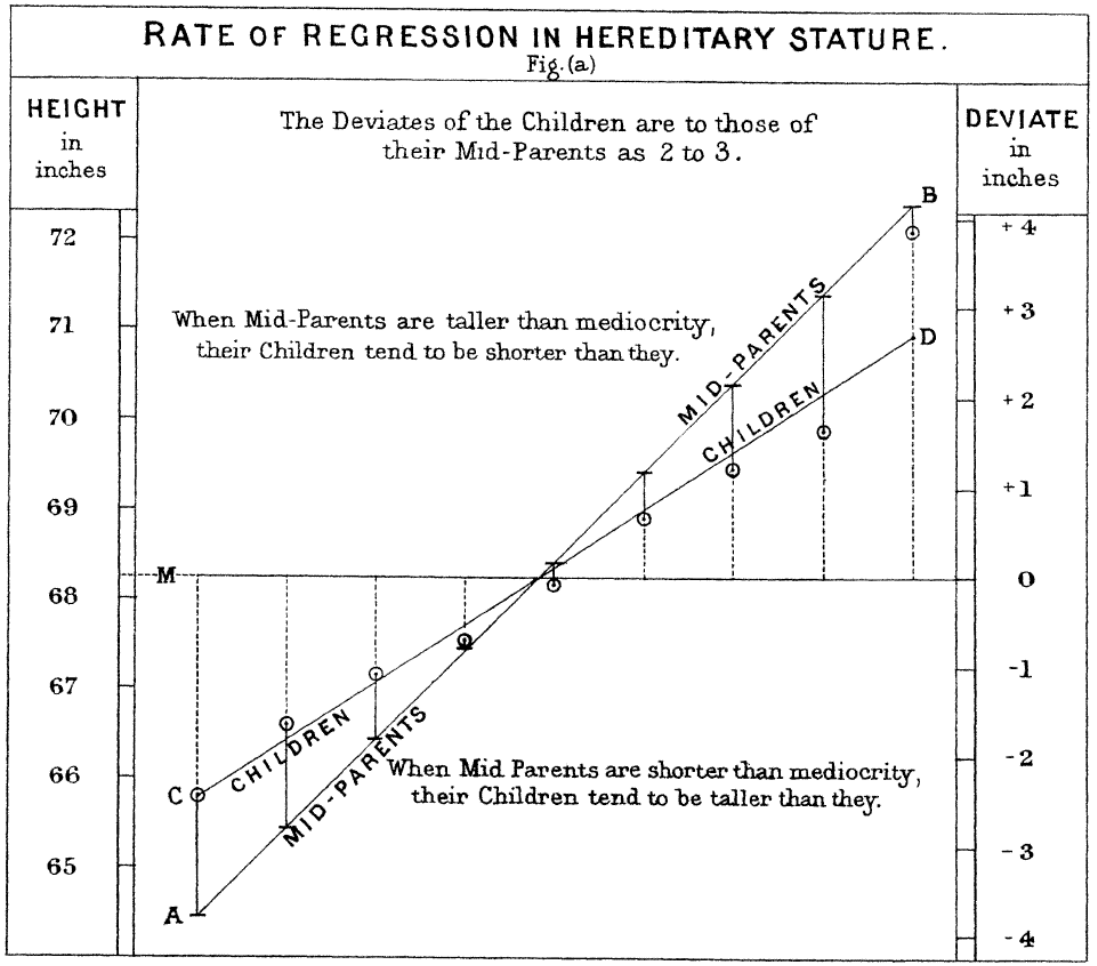
Regression *towards* Mediocrity *in* Hereditary Stature.

By Francis Galton, F.R.S., &c.

[With Plates IX and X.]

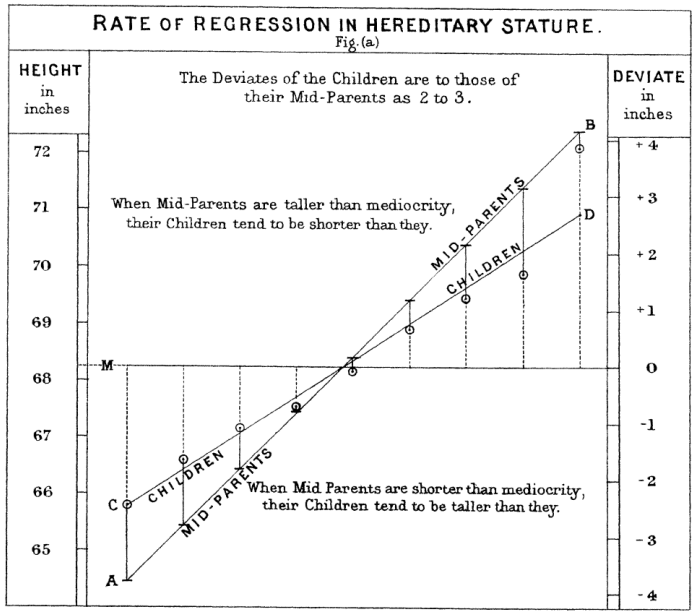
This memoir contains the data upon which the remarks on the Law of Regression were founded, that I made in my Presidential Address to Section H, at Aberdeen. That address, which will appear in due course in the Journal of the British Association, has already been published in "Nature," September 24th. I reproduce here the portion of it which bears upon regression, together with some amplification where brevity had rendered it obscure, and I have added copies of the diagrams suspended at the meeting, without which the letterpress is necessarily difficult to follow. My object is to place beyond doubt the existence of a simple and far-reaching law that governs the hereditary transmission of, I believe, every one of those simple qualities which all possess, though in unequal degrees. I once before ventured to draw attention to this law on far more slender evidence than I now possess.

Research question To what degree does parent height explain the variability of height of their children?



Galton 1886

Regression



$$\widehat{child.height} = \beta_0 + \beta_1 \cdot parent.height$$

- $\widehat{child.height}$ is the predicted or expected height of children given the predictor, $parent.height$.
- Simple regression uses the least squares method to estimate the parameters.
- Average height of children given the height of parents. For every one inch added to parent's height, child height will increase at $\beta_1 \cdot parent.height$.
- The intercept or constant, β_0, is the average predicted height of children if parent height was zero. This is not realistic, but can become more informative using transformation (centering).

Gelman & Hill 2007; Galton 1886

Regression

$$\widehat{child.height} = \beta_0 + \beta_1 \cdot parent.height$$

```
> summary(lm_m1)
Call:
lm(formula = child_height ~ parent_height, data = height_df)

Residuals:
    Min      1Q  Median      3Q     Max
-7.3137 -1.4399  0.0151  1.2638  6.9026

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   32.04498    5.65439   5.667 5.16e-08 ***
parent_height  0.52250    0.08478   6.163 4.02e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.194 on 195 degrees of freedom
Multiple R-squared:  0.163,         Adjusted R-squared:  0.1587
F-statistic: 37.98 on 1 and 195 DF,  p-value: 4.022e-09
```

- $\widehat{child.height}$ is the predicted or expected height of children given the predictor, $parent_height$.
- For every one inch added to parents' average height, child height will increase by 0.52 inches.
- The intercept or constant, β_0, is the average predicted height of children if parent height was zero (32 inches≈81 cm).

- **Residuals** $r_i = y_i - \hat{y}_i = y_i - (\beta_0 + \beta_1 x_i)$
- **Estimate** expected values
- **Standard error** uncertainty in estimates.
- **t-value** $Estimate/SE$
- **P(>t)** probability of observing a more extreme *t*-value upon repetition.
- **Residual error** the model error term (unexplained error)
- Degrees of freedom observations minus number of estimated parameters *195 = 197 – 2*.
- **R^2** A coefficient of determination. Proportion of variation (0-1) in *Y* explained by the model. We will talk about adjusted R^2 and the likes later (goodness-of-fit).
- **F-statistic and *p*-value** relates to strength of model relationship.

Regression

$$\widehat{child.height} = \beta_0 + \beta_1 \cdot parent.height$$

```
library(arm)
> display(lm_m1)
lm(formula = child_height ~ parent_height, data = height_df)
               coef.est coef.se
(Intercept)    32.04     5.65
parent_height   0.52     0.08
---
n = 197, k = 2
residual sd = 2.19, R-Squared = 0.16
```

n: observations, k: estimated parameters

Gelman *et al*. 2024 – arm vignette; Gelman & Hill 2007

Assumptions of linear regression in decreasing order of importance

I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

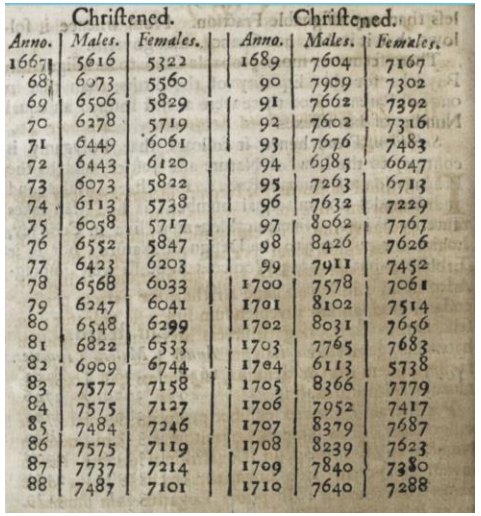
Gelman & Hill 2007

Assumptions of linear regression in decreasing order of importance: Galton height data 1886

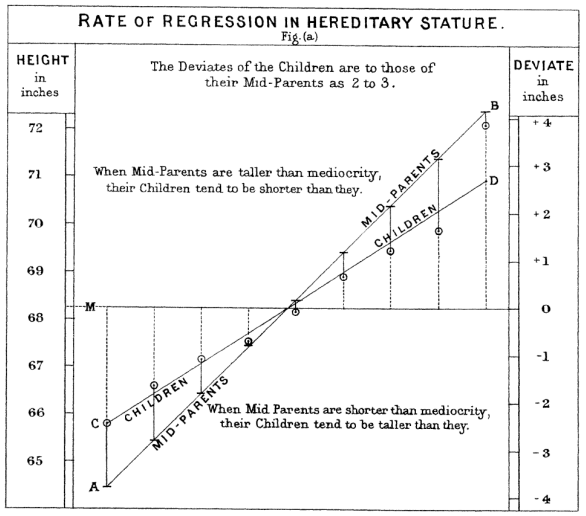
I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

Research question (RQ) To what degree does parent height explain the variability of height of their children?

- Data reflects RQ.
- Validity of outcome measure.
- Be able to generalise to the case to which the model will be applied.



Christened.			Christened.		
Anno.	Males.	Females.	Anno.	Males.	Females.
1667	5616	5322	1689	7604	7167
68	6073	5560	90	7909	7302
69	6506	5829	91	7662	7392
70	6278	5719	92	7602	7316
71	6449	6061	93	7676	7483
72	6443	6120	94	6985	6647
73	6073	5822	95	7263	6713
74	6113	5738	96	7632	7229
75	6058	5717	97	8062	7767
76	6552	5847	98	8426	7626
77	6423	6203	99	7911	7452
78	6568	6033	1700	7578	7061
79	6247	6041	1701	8102	7514
80	6548	6299	1702	8031	7656
81	6822	6533	1703	7765	7683
82	6909	6744	1704	6113	5738
83	7577	7158	1705	8366	7779
84	7575	7127	1706	7952	7417
85	7484	7246	1707	8379	7687
86	7575	7119	1708	8239	7623
87	7737	7214	1709	7840	7380
88	7487	7101	1710	7640	7288

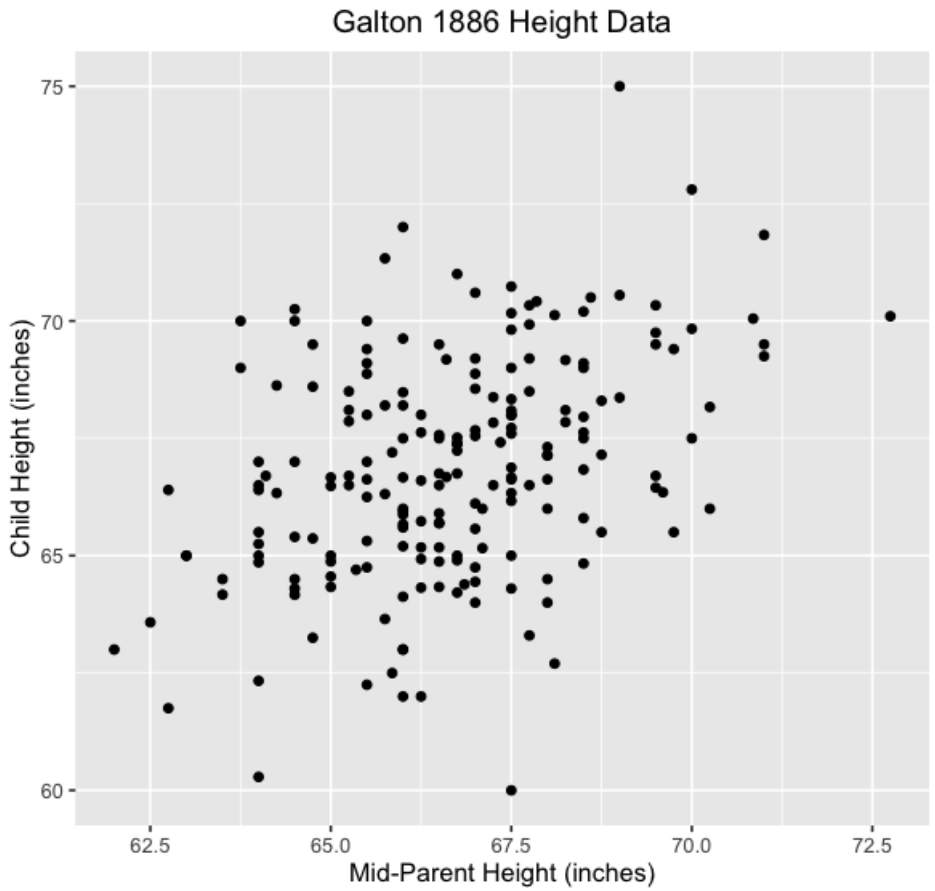


Gelman & Hill 2007

Assumptions of linear regression in decreasing order of importance: Galton height data 1886

I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

- The deterministic component of the model is linear $y = x \cdot \beta$.
- If violated, you could try transforming (*log*) the data or add interaction effects.

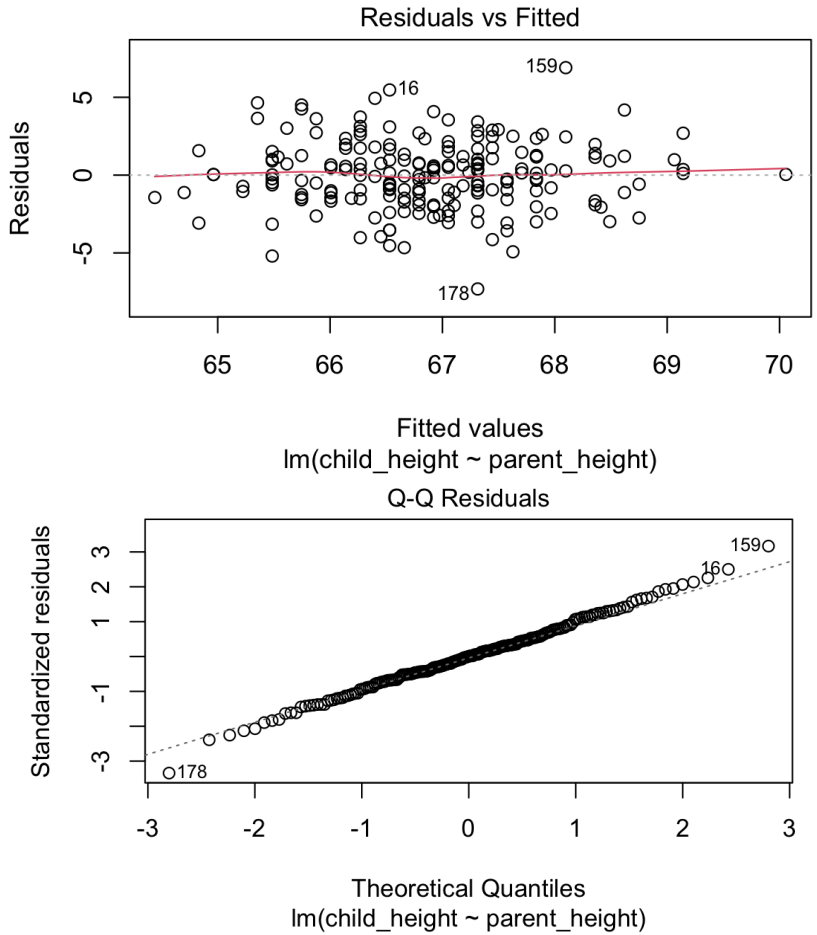


Gelman & Hill 2007

Assumptions of linear regression in decreasing order of importance: Galton height data 1886

I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

```
plot(model_fit)
```

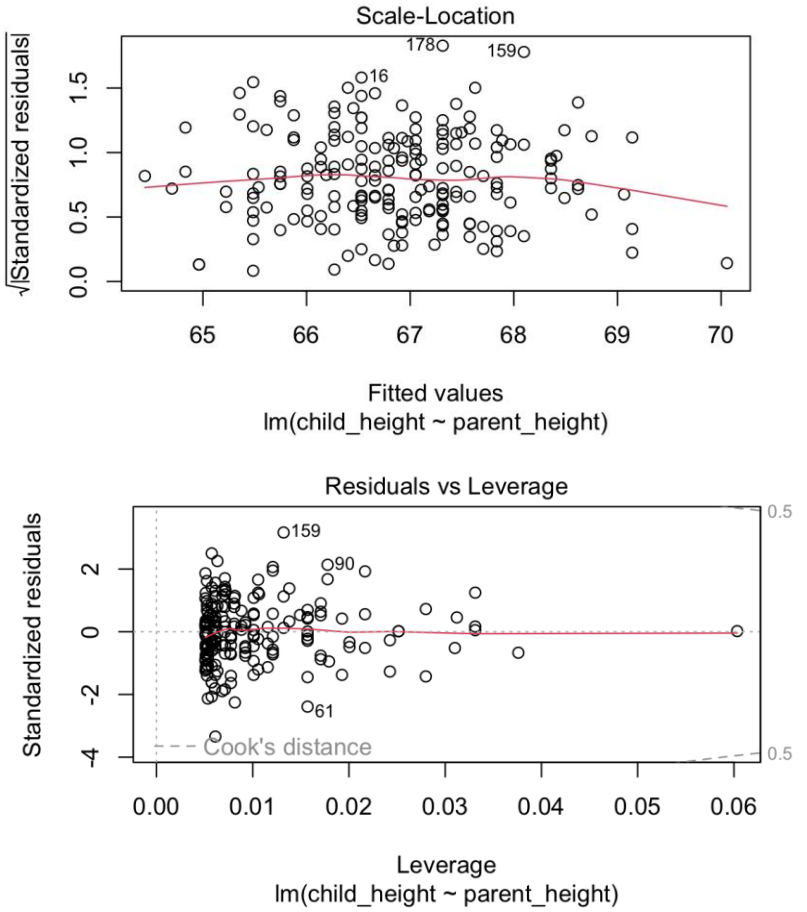


Gelman & Hill 2007

Assumptions of linear regression in decreasing order of importance: Galton height data 1886

I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

```
plot(model_fit)
```



Gelman & Hill 2007

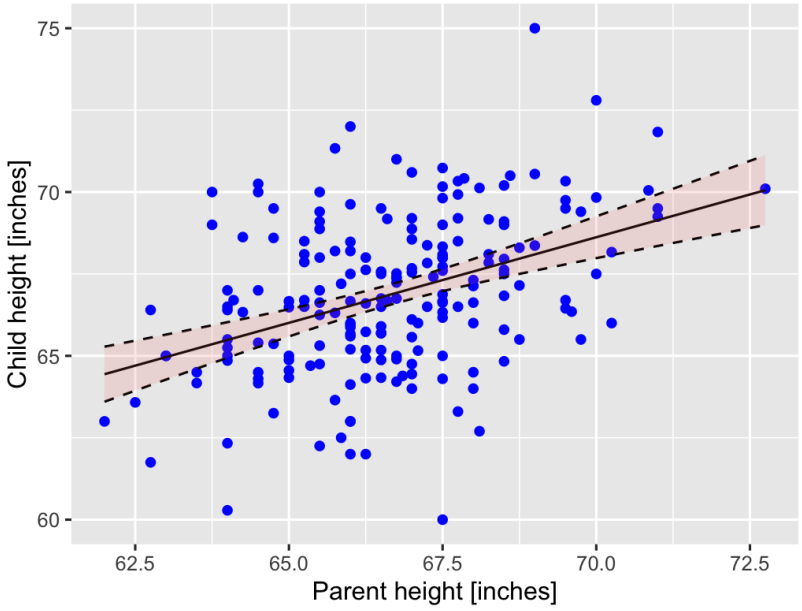
Prediction

```
lm(formula = child_height ~ parent_height, data = height_df)
               coef.est coef.se
(Intercept)    32.04     5.65
parent_height   0.52     0.08
---
n = 197, k = 2
residual sd = 2.19, R-Squared = 0.16
```

```
predict(model_fit, data, interval = c("confidence","prediction", level = 0.95)
```

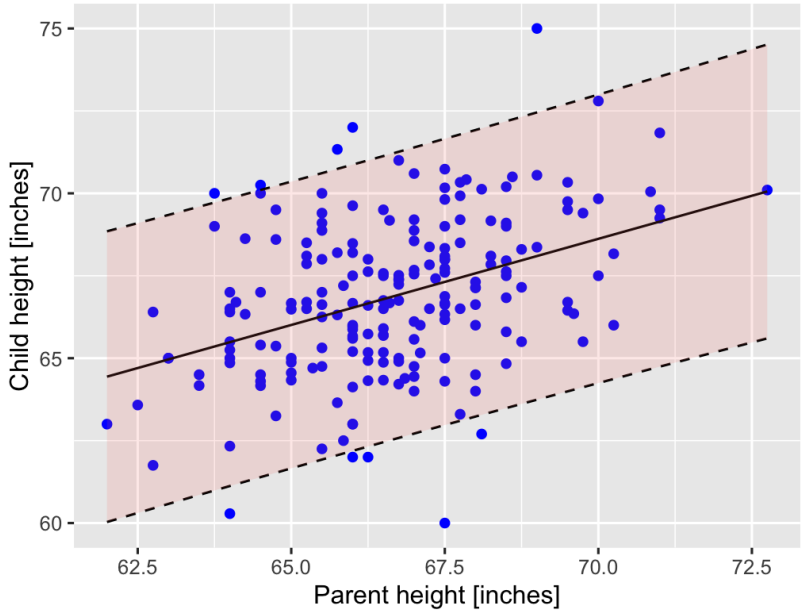
Prediction

Predicting a vector of $\tilde{n}$ outcomes given an $\tilde{n} \times k$ matrix of predictors, $\tilde{X}$. Then, $\tilde{y}_i$ is the outcome measure for the new data. The prediction is $\tilde{X}\hat{\beta}$ with a standard error slightly larger than $\hat{\sigma}$.



Confidence interval

- Upon repetition of an experiment the true population mean would fall within the 95% confidence interval of the mean 95% of the times.
- Relates to uncertainty in parameter estimates.
- Smallest at the x-value with the most information.



Prediction interval

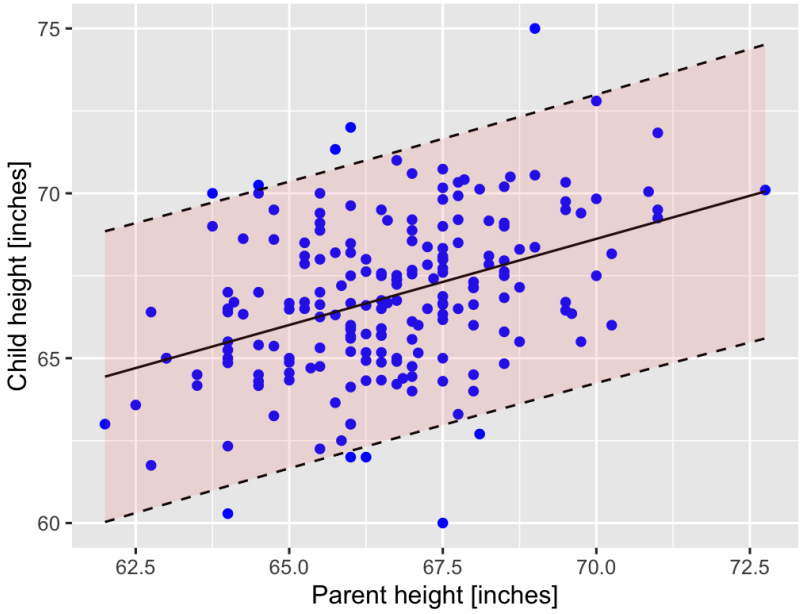
- Probability (*e.g.*, 95%) that future observations will fall within the interval.
- Relates to the prediction of future population samples / observations.

Kümmel *et al*. 2018; Gelman & Hill 2007

On the interpretation of the *Beta* coefficients

- The *Beta*, correlation coefficient, is a measure of the strength of a linear relationship.
- **The predictive interpretation** average difference between two groups were *Beta* differs.
- **The counterfactual interpretation** the expected change in *y* given a change in *Beta*, for example on an individual level (causal inference). This requires more assumptions.
- A high Beta does not imply causality.
- The beta coefficient does not say anything about the curvilinear relationship (*r*) between *x* and *y*.
- Always plot the data before estimation.

Larsen & Marx 2006; Gelman & Hill 2007



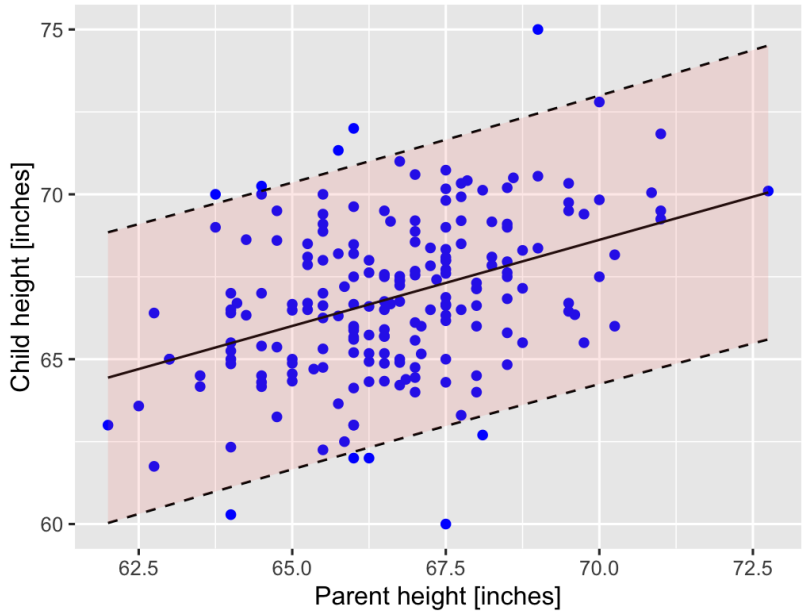
ANTHROPOLOGICAL MISCELLANEA.

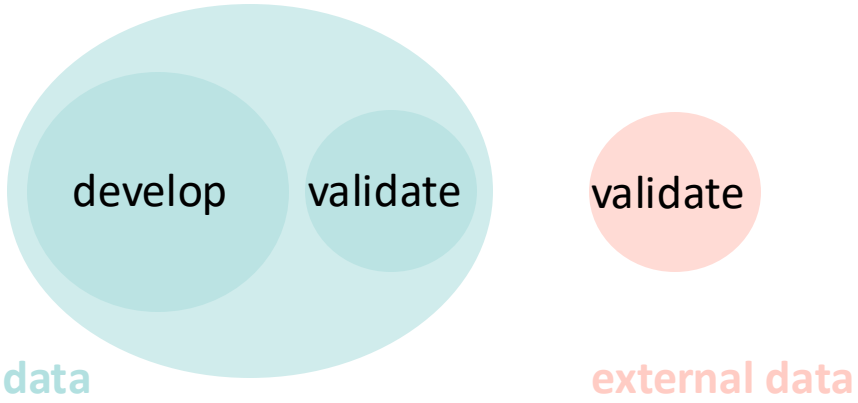
REGRESSION *towards* MEDIOCRITY *in* HEREDITARY STATURE.
By FRANCIS GALTON, F.R.S., &c.

[WITH PLATES IX AND X.]

THIS memoir contains the data upon which the remarks on the Law of Regression were founded, that I made in my Presidential Address to Section H, at Aberdeen. That address, which will appear in due course in the Journal of the British Association, has already been published in "Nature," September 24th. I reproduce here the portion of it which bears upon regression, together with some amplification where brevity had rendered it obscure, and I have added copies of the diagrams suspended at the meeting, without which the letterpress is necessarily difficult to follow. My object is to place beyond doubt the existence of a simple and far-reaching law that governs the hereditary transmission of, I believe, every one of those simple qualities which all possess, though in unequal degrees. I once before ventured to draw attention to this law on far more slender evidence than I now possess.

External validation





Kümmel *et al*. 2018; Gelman & Hill 2007

L2-4 The method of least-squares

The method of least squares

Carl Friedrich Guass & Adrien-Marie Legendre

Minimise sum of squares of squares (*L*), the vertical distances from the data to a presumed polynomial (*p*):

$$L = \sum_{i=1}^{n} [y_i - p(x_i)]^2$$

Linear polynomial ($a + bx_i$), minimise coefficeints:

$$L = \sum_{i=1}^{n} [y_i - (\beta_0 + \beta_1 x_i)]^2$$

The method of least squares

A RELATION BETWEEN DISTANCE AND RADIAL VELOCITY AMONG EXTRA-GALACTIC NEBULAE

By Edwin Hubble

Mount Wilson Observatory, Carnegie Institution of Washington

Communicated January 17, 1929

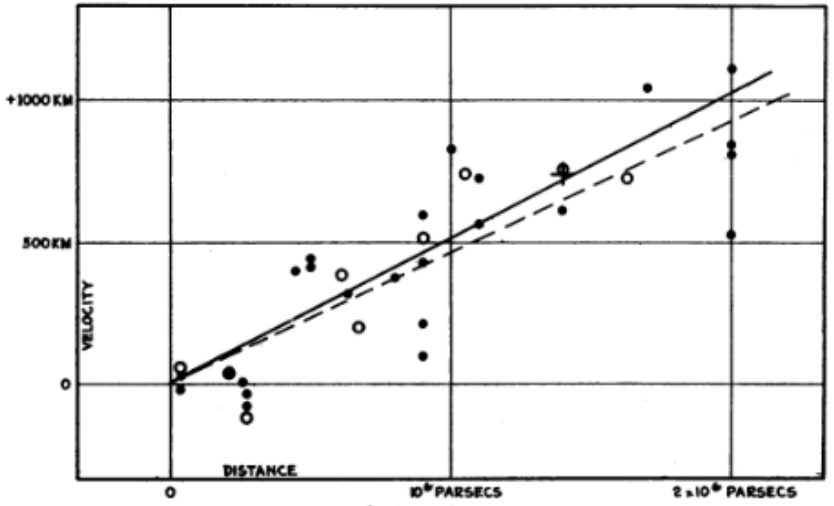
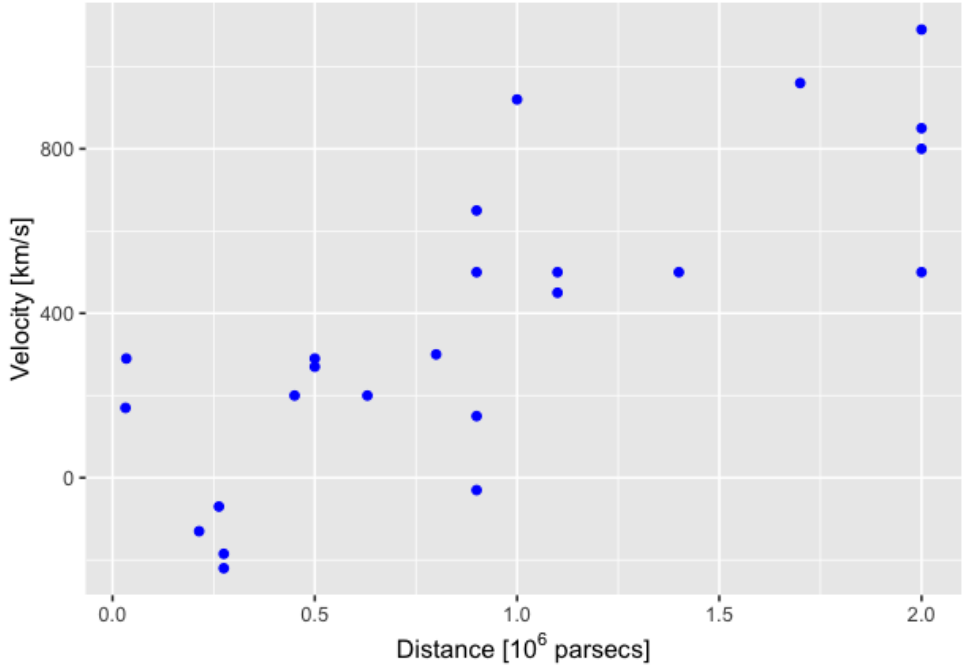


FIGURE 1

Velocity-Distance Relation among Extra-Galactic Nebulae.

Radial velocities, corrected for solar motion, are plotted against distances estimated from involved stars and mean luminosities of nebulae in a cluster. The black discs and full line represent the solution for solar motion using the nebulae individually; the circles and broken line represent the solution combining the nebulae into groups; the cross represents the mean velocity corresponding to the mean distance of 22 nebulae whose distances could not be estimated individually.



The method of least squares

Minimise sum of squares of squares (L), the vertical distances from the data to a presumed polynomial (p):

$$L = \sum_{i=1}^{n} [y_i - p(x_i)]^2$$

Linear polynomial ($a + bx_i$), minimise coefficeints:

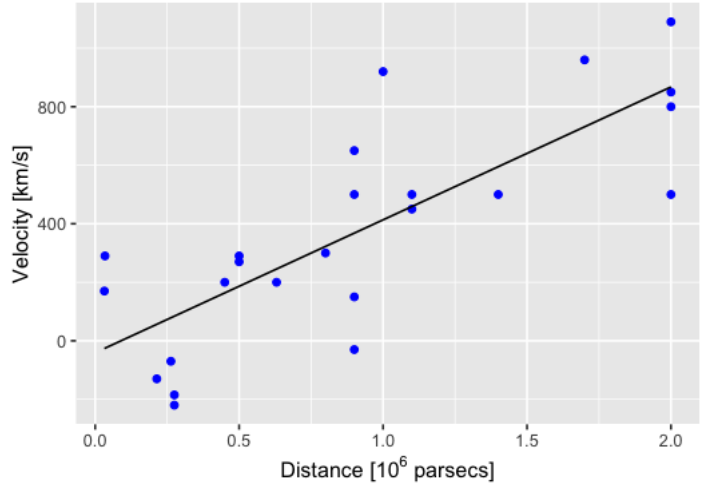
$$L = \sum_{i=1}^{n} [y_i - (\beta_0 + \beta_1 x_i)]^2$$

Slope:

$$\beta_1 = \frac{\sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n(\sum_{i=1}^{n} x_i^2) - (\sum_{i=1}^{n} x_i)}$$

Intercept:

$$\beta_0 = \frac{\sum_{i=1}^{n} y_i - b \sum_{i=1}^{n} x_i}{n} = \bar{y} - \beta_1 \bar{x}$$




```
# calculating slope using the method of least squares - let's
create a function:
lm_beta1 <- function(x, y, n){
  ( n*sum(x*y) - sum(x)*sum(y) ) / ( n*sum(x^2) - sum(x)^2 )
}

beta1 <- lm_beta1(hubble_dat$Distance, hubble_dat$Velocity,
length(hubble_dat$Cluster))
```

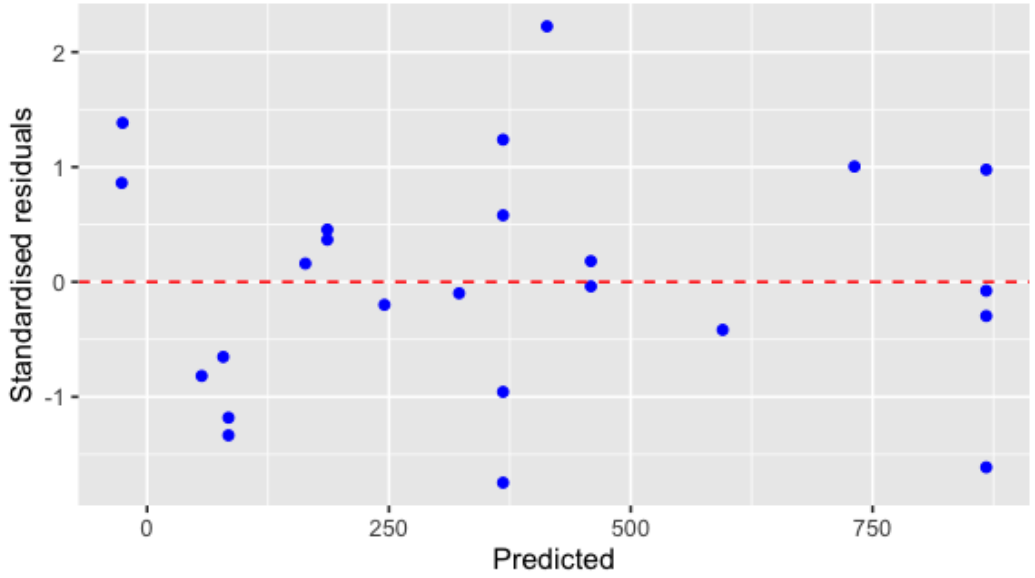
Parameter	Value
Intercept	-40.78
Slope	454.16

Residual errors

The difference in observed y_i and the least squares line when $x = x_i$ is the ith residual (e_i). It's magnitude is the failure of the least squares line to model the datapoint.

$$e_i = y_i - \hat{y}_i = y_i - (a + bx_i)$$

A graph of e_i versus x_i for all i, is the residual plot.



$$SE_e = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{df}} = 232.9$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 0.624$$

Standard errors of estimates, t-ratios and *p*-values

Standard errors

Slope:

$$SE_{\beta_1} = \sqrt{\frac{\sum_{i=1}^{n} e_i^2}{n-2} \Big/ \sum_{i=1}^{n}(x_i - \bar{x})^2} = 75.24$$

Intercept:

$$SE_{\beta_0} = \sqrt{\frac{\sum_{i=1}^{n} e_i^2}{n-2}} \cdot \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = 83.44$$

$$t - statistic = \frac{\hat{\beta}}{SE_{\beta}}, \qquad t_{\beta 1} = 6.04, \qquad t_{\beta 0} = -0.49$$

H_0: $\beta_j = 0$

H_1: $\beta_j \neq 0$

```
p_b1 <- 2 * pt(abs(t_b1), df_p,
lower.tail = FALSE)
>
p_b1: 4.48e-06
p_b0: 0.630
```

f-statistic for the model

H_0: $\beta_1 = \beta_j = 0$

H_1: $\beta_j \neq 0 \; for \; at \; least \; on \; value \; of \; j$

$$f - statistic = \frac{mean \; of \; squares \; of \; model}{mean \; square \; error} = 36.44$$

```
m1_f_p <- pf(fstat, df_1, df, lower.tail =
FALSE)
>
m1_f_p: 4.48e-06
```

Verifying against the built-in R function

```
Call:
lm(formula = Velocity ~ Distance, data = hubble_dat)

Residuals:
    Min      1Q  Median      3Q     Max
-397.96 -158.10  -13.16  148.09  506.63

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -40.78      83.44  -0.489     0.63
Distance      454.16      75.24   6.036 4.48e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 232.9 on 22 degrees of freedom
Multiple R-squared:  0.6235,      Adjusted R-squared:  0.6064
F-statistic: 36.44 on 1 and 22 DF,  p-value: 4.477e-06
```

Parameter	Value	SE	t-value	Pr(>|t|)
Intercept	-40.78	83.44	-0.49	0.630
Slope	454.16	75.24	6.04	4.48e-06
Residual SE	232.9			
R^2	0.624			
F-statistic	36.44			
p-value	4.48e-06			

Multiple linear regression

$$\boldsymbol{X}\beta = y$$

$$X = \begin{bmatrix} x_{1,1} & \cdots & x_{1,p} \\ \vdots & \ddots & \vdots \\ x_{n,1} & \cdots & x_{n,p} \end{bmatrix} \qquad \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \qquad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Objective function:

$$\hat{\beta} = \arg\min S(\beta) = \sum_{i=1}^{n} \left| y_i - \sum_{j=1}^{p} X_{ij}\beta_j \right|^2 = \|y - \boldsymbol{X}\beta\|^2$$

$$(\boldsymbol{X}^T\boldsymbol{X})\hat{\beta} = \boldsymbol{X}^T y$$

$$\hat{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T y$$

Least squares variations

Minimise sum of squares of squares (L), the vertical distances from the data to a presumed model (m):

$$L = \sum_{i=1}^{n} [y_i - m(x_i)]^2$$

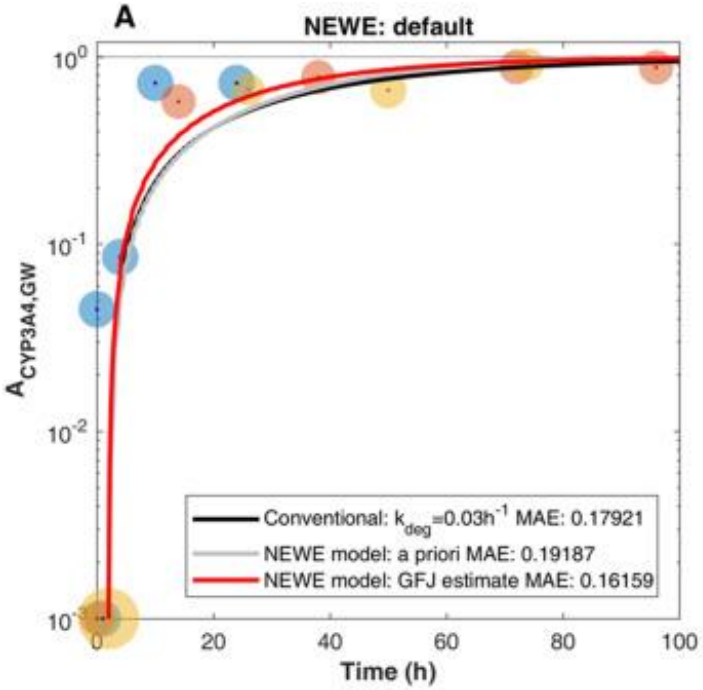
Weighted least squares:

$$S = \sum_{i=1}^{n} [W_{ii}(y_i - m(x_i))]^2, \qquad W_{ii} = \frac{1}{\sigma_i^2}$$

```
m1 <- lm(y ~ x, data, weights = c(w1,w2, …, wii))
```

In case of heteroscedasticity, weights by variance of observations.

Generalised least squares, nonlinear least squares.

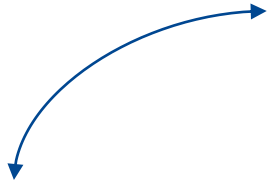


Darwich *et al*. 2019



L2-5 Data Formatting

Wide data



Y	*X*	*Z*
y_1	x_1	z_1
y_2	x_2	z_2

Y	*X*
y_1	x_1
y_2	x_2
y_3	x_3

Wide data

ID	BW	1-weeks	3-weeks	6-weeks
1	75	55	67	96
2	86	67	70	88
3	68	45	60	78
4	89	57	71	94
5	77	68	72	105
6	80	51	45	75

```
t.test( (6-weeks - 1-weeks),
paired = TRUE…)
```

Wide data

ID	BW	T1-weeks	T3-weeks	T6-weeks	C1-weeks	C3-weeks	C6-weeks
1	75	55	67	96	54	57	49
2	86	67	70	88	60	71	55
3	68	45	60	78	41	51	45
4	89	57	71	94	51	56	49
5	77	68	72	105	55	61	63
6	80	51	45	75	45	50	51

```
t.test( (T6-weeks – C6-weeks),
paired = TRUE…)
```

Long data

ID	BW	T1-weeks	T3-weeks	T6-weeks	C1-weeks	C3-weeks	C6-weeks
1	75	55	67	96	54	57	49
2	86	67	70	88	60	71	55
3	68	45	60	78	41	51	45
4	89	57	71	94	51	56	49
5	77	68	72	105	55	61	63
6	80	51	45	75	45	50	51

ID	BW	Treatment	Time	DV
1	75	0	1	54
1	75	0	3	57
1	75	0	6	49
1	75	1	1	55
1	75	1	3	67
1	75	1	6	96
2	86	0	1	60
2	86	0	3	71
…				

```
m1 <- lm(DV ~ TIME, data)
barplot(DV  ~ Treatment, data)
```

Converting from wide to long data in R

```
# example: http://www.cookbook-
r.com/Manipulating_data/Converting_data_between_wide_and_long_format/
# load library for data formatting
library(tidyr)

# convert ID to factor
w_dat$ID <- factor(w_dat$ID)

# convert to long data
l_dat <- gather(w_dat, time, DV, Treat1w:Control6w, factor_key=TRUE)

# some additional manipulations
l_dat$Time <- with(l_dat, ifelse(grepl("3", time), 3, ifelse(grepl("6", time), 6,
1)))
l_dat$Treat <- with(l_dat, ifelse(grepl("Treat", time), 1, 0))

# remove column
l_dat <- subset(l_dat, select = -time)

# sort by columns
l_dat[ with(l_dat, order(ID, Treat, Time)),]

# we might also want to rearrange order of columns
library(tibble)
l_dat <- l_dat[, c("ID","BW","Treat","Time","DV")]
```

Dummy variables

ID	BW	Sex
1	75	Female
1	75	Female
1	75	Female
1	75	Female
1	75	Female
1	75	Female
2	86	Male
2	86	Male
…		

ID	BW	Sex_female	Sex_male
1	75	1	0
1	75	1	0
1	75	1	0
1	75	1	0
1	75	1	0
1	75	1	0
2	86	0	1
2	86	0	1
…			

ID	BW	Sex_female
1	75	1
1	75	1
1	75	1
1	75	1
1	75	1
1	75	1
2	86	0
2	86	0
…		

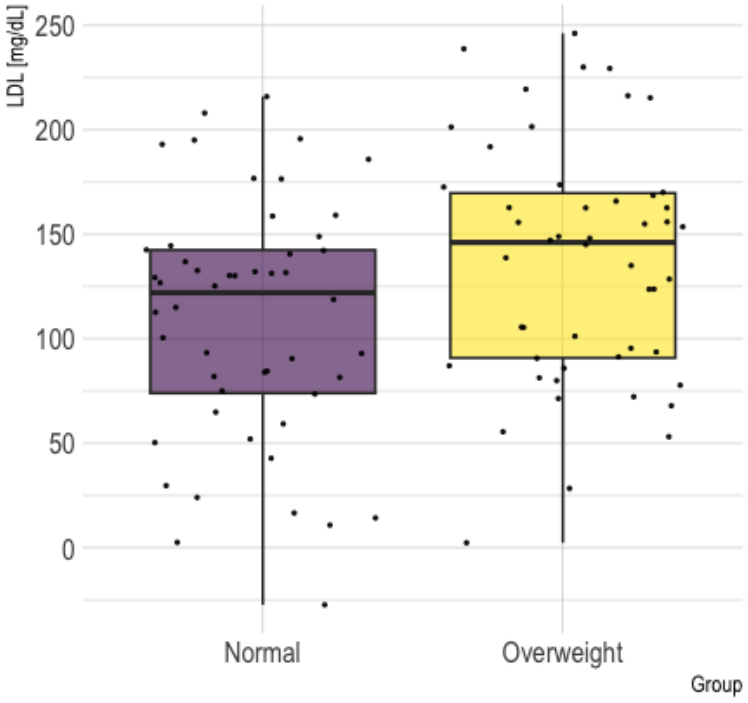
```
library(fastDummies)
l_dat <- dummy_cols(l_dat, select_columns = ”Sex”)
```

Often work with $n_{categories}$ – 1 dummy variables (one group will act reference):

```
l_dat <- subset(l_dat, select =-Sex_male)
```

L2-6 Linear regression, the t-test and ANOVA

The t-test and linear regression



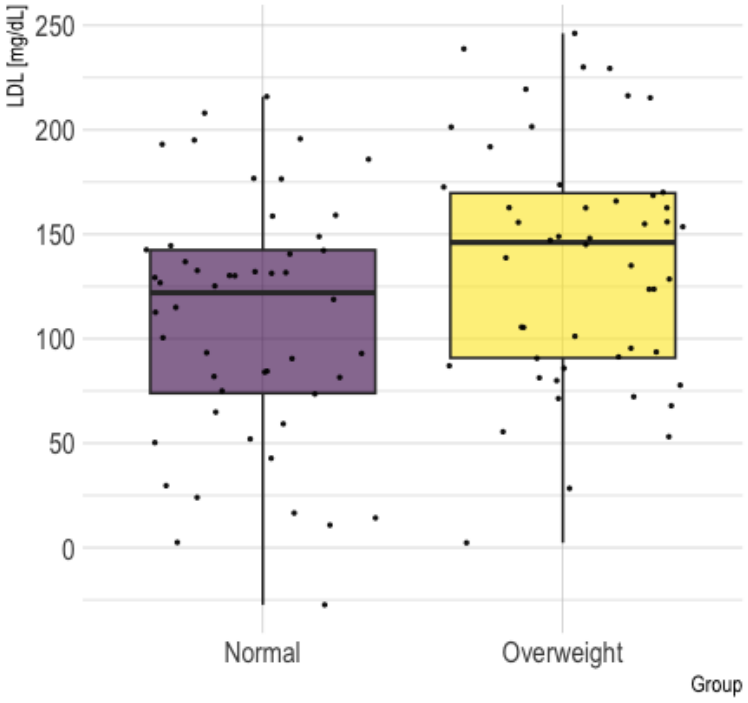
Open Access

Original Article

Correlation between Body Mass Index and Lipid Profile in patients with Type 2 Diabetes attending a tertiary care hospital in Peshawar

Arshad Hussain[1], Iftikhar Ali[2], Waqar Ahmad Kaleem[3], Fatima Yasmeen[4]

The t-test and linear regression



$$t = \frac{difference\ in\ means}{standard\ error}$$

t-test

$H_0: (\mu_A - \mu_B) = 0$
$H_1: (\mu_A - \mu_B) \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_p^2}{n_1} + \frac{\sigma_p^2}{n_2}}}$$

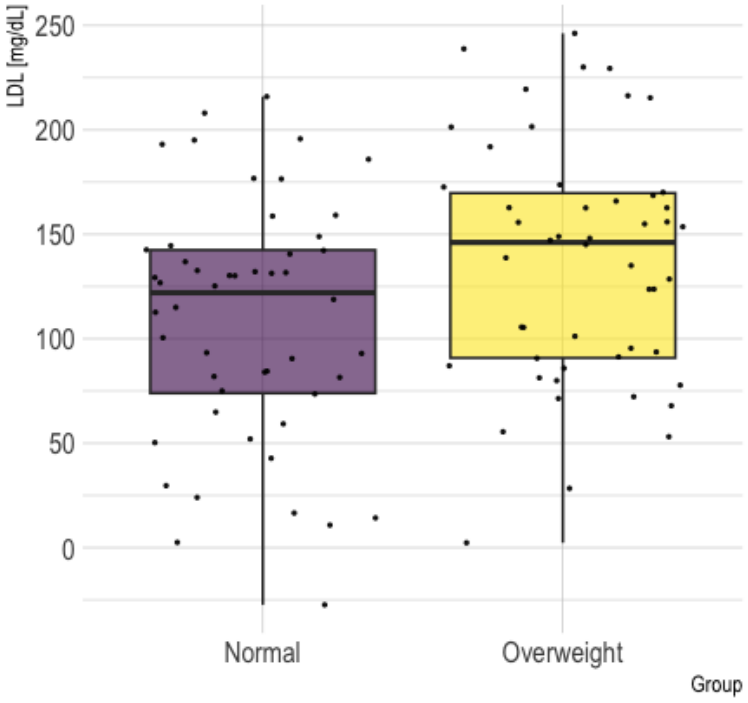
linear regression

$H_0: \beta_j = 0$
$H_1: \beta_j \neq 0$

$$t = \frac{\hat{\beta}_j}{SE_{\beta,j}}$$

$$\widehat{LDL} = \hat{\beta}_0 + \hat{\beta}_1 \cdot Overweight$$

The t-test and linear regression



$$t = \frac{difference\ in\ means}{standard\ error}$$

t-test

$H_0: (\mu_A - \mu_B) = 0$
$H_1: (\mu_A - \mu_B) \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_p^2}{n_1} + \frac{\sigma_p^2}{n_2}}}$$

linear regression

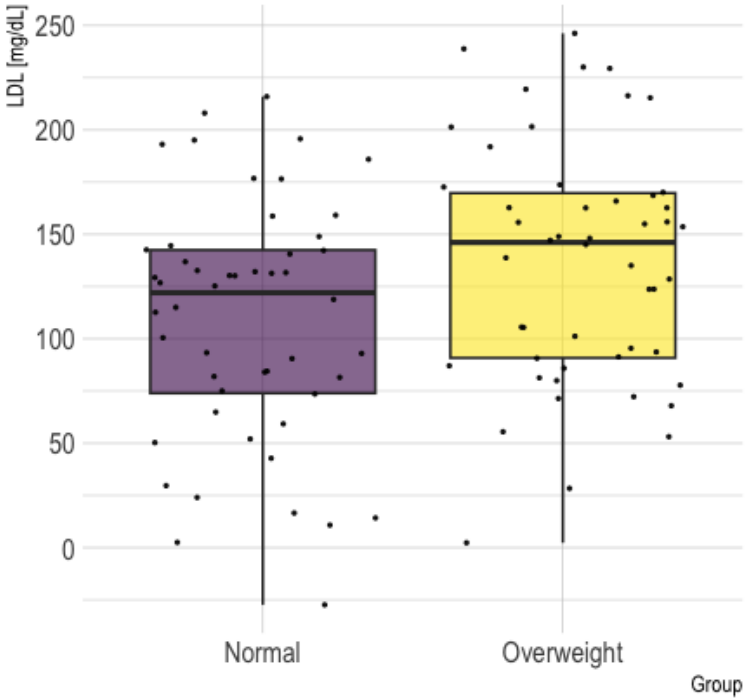
$H_0: \beta_j = 0$
$H_1: \beta_j \neq 0$

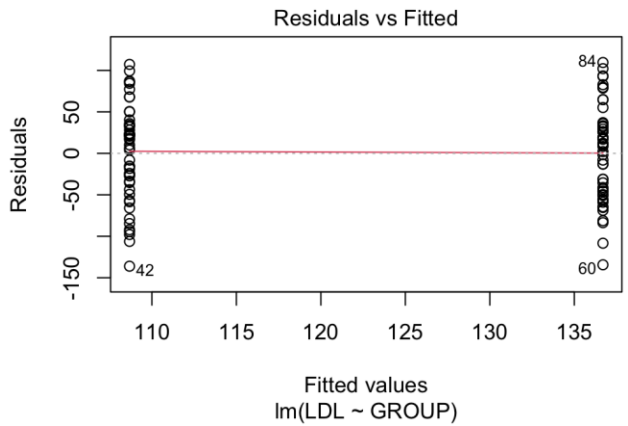
$$t = \frac{\hat{\beta}_j}{SE_{\beta,j}}$$

```
        Welch Two Sample t-test

data:  ldl_overweight and ldl_bwnormal
t = 2.417, df = 97.987, p-value = 0.0175
alternative hypothesis: true difference
in means is not equal to 0
95 percent confidence interval:
  5.014048 51.025409
sample estimates:
mean of x mean of y
 136.6955  108.6757
```

The t-test and linear regression





$$t = \frac{difference\ in\ means}{standard\ error}$$

t-test

$H_0: (\mu_A - \mu_B) = 0$
$H_1: (\mu_A - \mu_B) \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_p^2}{n_1} + \frac{\sigma_p^2}{n_2}}}$$

```
        Welch Two Sample t-test

data:  ldl_overweight and ldl_bwnormal
t = 2.417, df = 97.987, p-value = 0.0175
alternative hypothesis: true difference
in means is not equal to 0
95 percent confidence interval:
  5.014048 51.025409
sample estimates:
mean of x mean of y
 136.6955  108.6757
```

linear regression

$H_0: \beta_j = 0$
$H_1: \beta_j \neq 0$

$$t = \frac{\hat{\beta}_j}{SE_{\beta,j}}$$

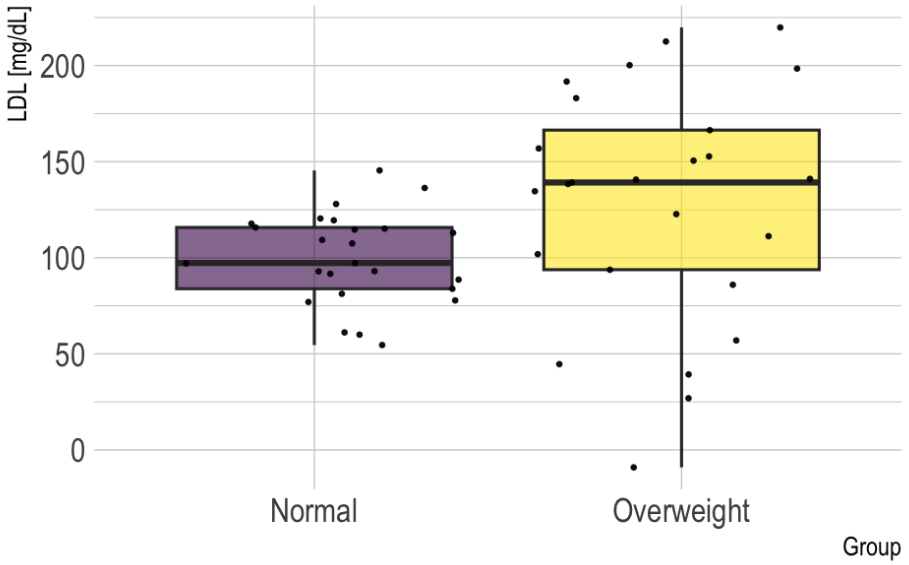
```
Call:
lm(formula = LDL ~ GROUP, data = ldl_dat)

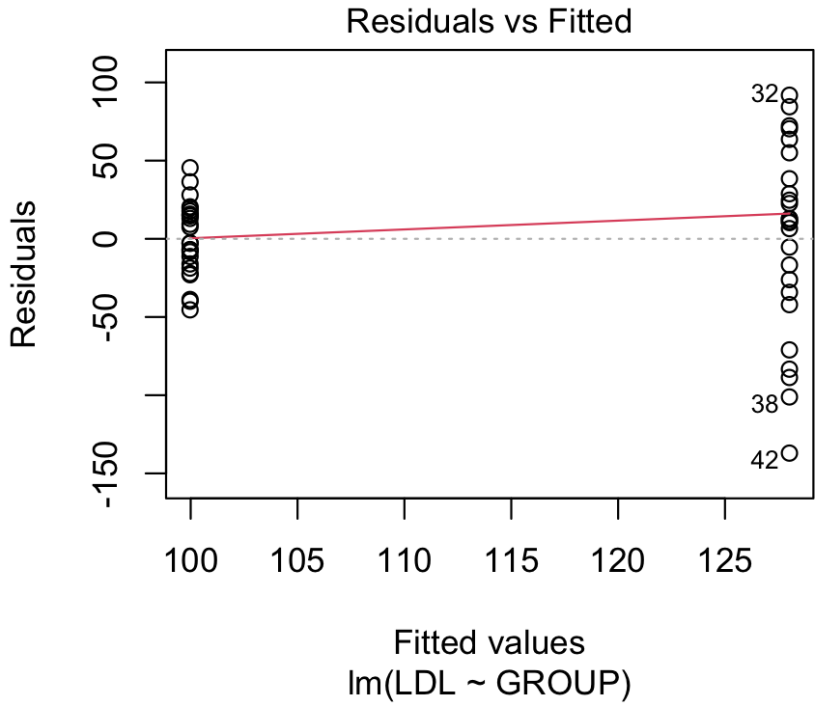
Residuals:
    Min      1Q  Median      3Q     Max
-136.09  -44.20   10.29   33.68  109.53

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  108.676      8.197  13.257   <2e-16 ***
GROUP         28.020     11.593   2.417   0.0175 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 57.96 on 98 degrees of freedom
Multiple R-squared:  0.05626,      Adjusted R-squared:  0.04663
F-statistic: 5.842 on 1 and 98 DF,  p-value: 0.0175
```

Generalised least squares for unequal variance



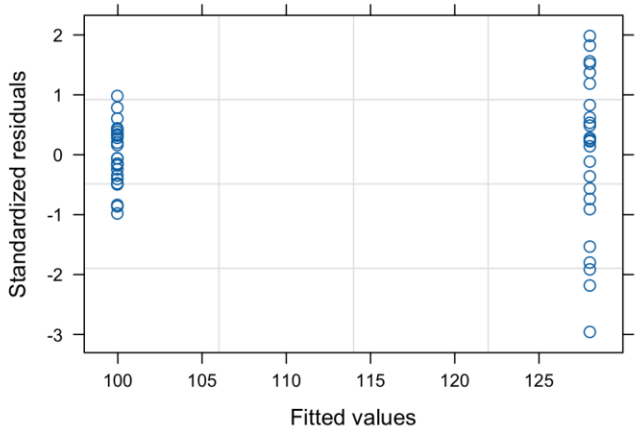



```
lm(LDL ~ GROUP, ldl2_dat)
```

Generalised least squares for unequal variance

- Unequal variance (heteroscedasticity) is not necessarily a problem, estimates of coefficients remain unbaised.
- If we are interested in model-based hypothesis testing, then we need to correct for this as standard errors and *p*-values can be affected.
- We can use generalised least squares `gls()` with weighting to account for the unequal variance. For example, `library(nlme)`.
- There are different models of variance, you can read more about them in the vignette, section Variance Function Classes (Pinheiro *et al*. 2025).

Generalised least squares for unequal variance



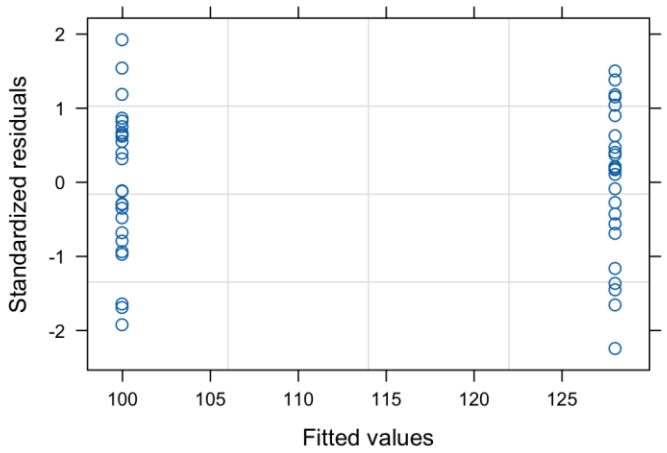

```
Generalized least squares fit by REML
  Model: LDL ~ GROUP
  Data: ldl_dat2
     AIC      BIC    logLik
  516.89 522.5036 -255.445

Coefficients:
               Value Std.Error  t-value p-value
(Intercept) 99.97057  9.265843 10.78915  0.0000
GROUP       28.04492 13.103880  2.14020  0.0374

 Correlation:
      (Intr)
GROUP -0.707

Standardized residuals:
       Min         Q1        Med         Q3        Max
-2.9578182 -0.4593343  0.1812318  0.4758684  1.9820364

Residual standard error: 46.32921
Degrees of freedom: 50 total; 48 residual
```




```
Generalized least squares fit by REML
  Model: LDL ~ GROUP
  Data: ldl_dat2
       AIC      BIC    logLik
  499.8613 507.3461 -245.9306

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | GROUP
 Parameter estimates:
       0        1
1.000000 2.586388

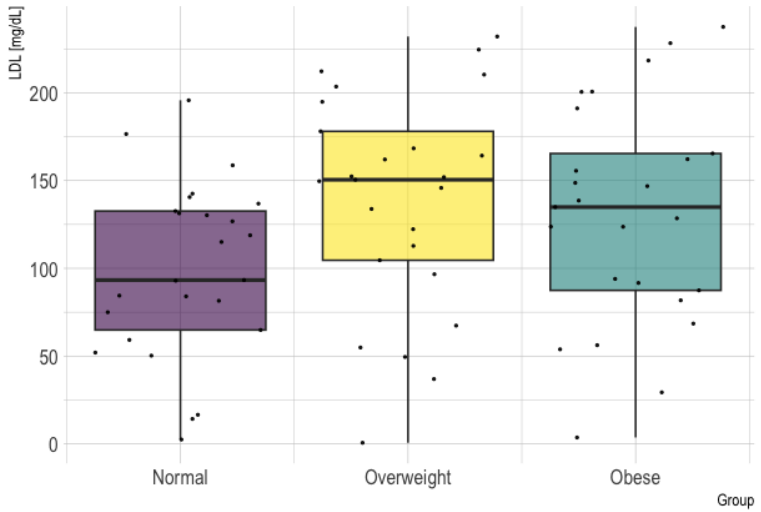
Coefficients:
               Value Std.Error   t-value p-value
(Intercept) 99.97057  4.725566 21.155257  0.0000
GROUP       28.04492 13.103888  2.140199  0.0374

 Correlation:
      (Intr)
GROUP -0.361

Standardized residuals:
       Min         Q1        Med         Q3        Max
-2.2423782 -0.6503673  0.1758984  0.6615324  1.9240968

Residual standard error: 23.62783
Degrees of freedom: 50 total; 48 residual
```

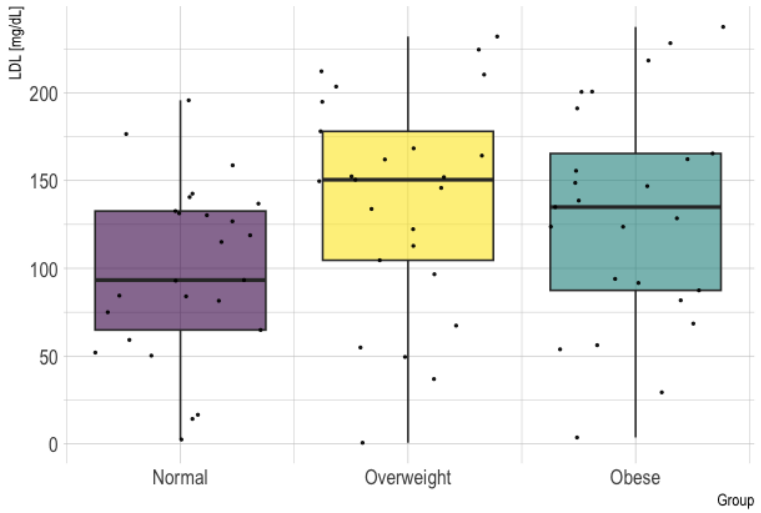
The ANOVA and linear regression



$H_0: \mu_1 = \mu_2 = \mu_3 = 0$
$H_1: \exists\ i, j \mid \mu_i \neq \mu_j$

$H_0: \beta_1 = \beta_2 = 0$
$H_1: \exists\ j \mid \beta_j \neq 0$

The ANOVA and linear regression

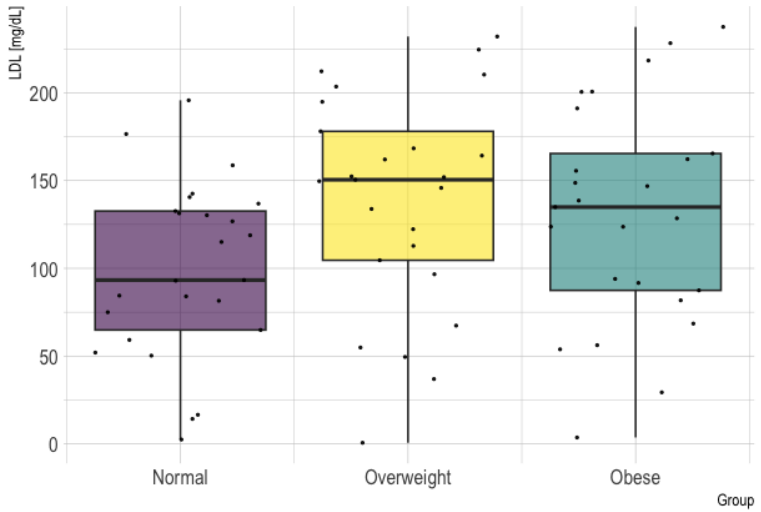



```
> anova_ldl3 <- aov(LDL ~ GROUPN, data = ldl3_dat2)
> summary(anova_ldl3)
             Df  Sum Sq Mean Sq F value Pr(>F)
GROUPN        2   22415   11207   3.271 0.0437 *
Residuals    72  246706    3426
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05
‘.’ 0.1 ‘ ’ 1
```

sum of squared errors of the full model and the null model.

```
> ldl3_m1 <- lm(LDL ~ Overweight + Obese, ldl3_dat2)
> summary(ldl3_m1)
Call:
lm(formula = LDL ~ Overweight + Obese, data =
ldl3_dat2)
Residuals:
     Min       1Q   Median       3Q      Max
-138.568  -39.510    7.742   36.172  106.685
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)     99.07      11.71   8.462  2.1e-12 ***
Overweight      40.13      16.56   2.424   0.0179 *
Obese           31.77      16.56   1.919   0.0589 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’
0.1 ‘ ’ 1
Residual standard error: 58.54 on 72 degrees of freedom
Multiple R-squared:  0.08329,     Adjusted R-squared:
0.05782
F-statistic: 3.271 on 2 and 72 DF,  p-value: 0.04369
```

The ANOVA and linear regression




```
> library(DescTools)
> PostHocTest(anova_ldl3, method ="lsd")
  Posthoc multiple comparisons of means : Fisher LSD
    95% family-wise confidence level

$GROUPN
                       diff     lwr.ci   upr.ci   pval
Obese-Normal       31.773256  -1.231547 64.77806 0.0589 .
Overweight-Normal  40.129848   7.125045 73.13465 0.0179 *
Overweight-Obese    8.356592 -24.648211 41.36139 0.6153

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Fisher LSD pairwise test (uncorrected).

```
> ldl3_m1 <- lm(LDL ~ Overweight + Obese, ldl3_dat2)
> summary(ldl3_m1)
Call:
lm(formula = LDL ~ Overweight + Obese, data =
ldl3_dat2)
Residuals:
     Min       1Q   Median       3Q      Max
-138.568  -39.510    7.742   36.172  106.685
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    99.07      11.71   8.462  2.1e-12 ***
Overweight     40.13      16.56   2.424   0.0179 *
Obese          31.77      16.56   1.919   0.0589 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
Residual standard error: 58.54 on 72 degrees of freedom
Multiple R-squared:  0.08329,     Adjusted R-squared:
0.05782
F-statistic: 3.271 on 2 and 72 DF,  p-value: 0.04369
```

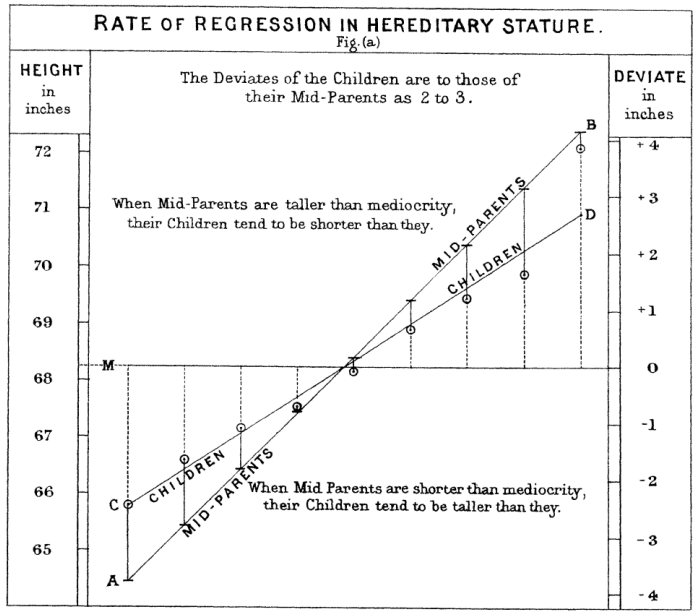
Regression models as alternatives to statistical tests

I. Producing estimates of effect size is not straightforward for rank tests or perumtation tests.
II. It is not straightforward to extend these models to more complex experiments (data structures), *e.g.*, repeated measures.
III. Regression is often more straightforward to adapt than stastical tests. For example, Wilcoxon-Mann-Whitney is a special case of proportional odds ordinal logistic regression, ANOVA and Kruskal-Wallis – regression models with dummy variables, etc.

Harrell 2015

L2-7 Multiple Linear Regression

Linear regression

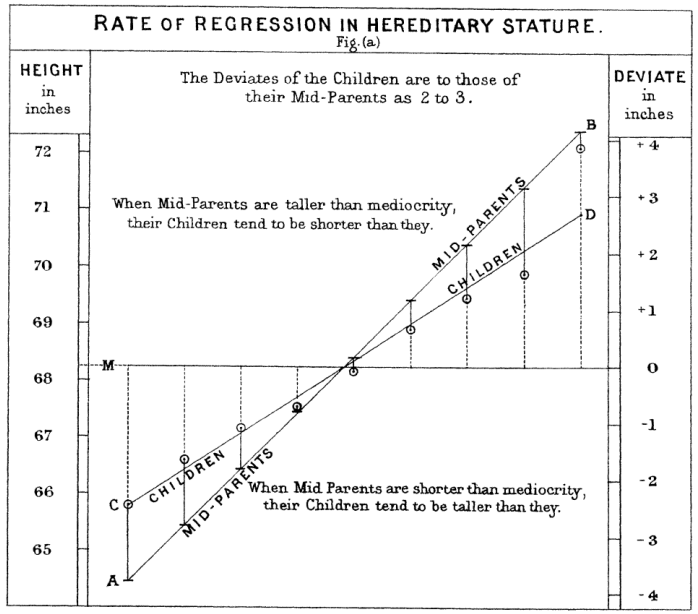


Research question To what degree does parent height explain the variability of height of their children?

$$\widehat{child.height} = \beta_0 + \beta_1 \cdot parent.height$$

$H_0: \beta_1 = 0$
$H_1: \beta_1 \neq 0$

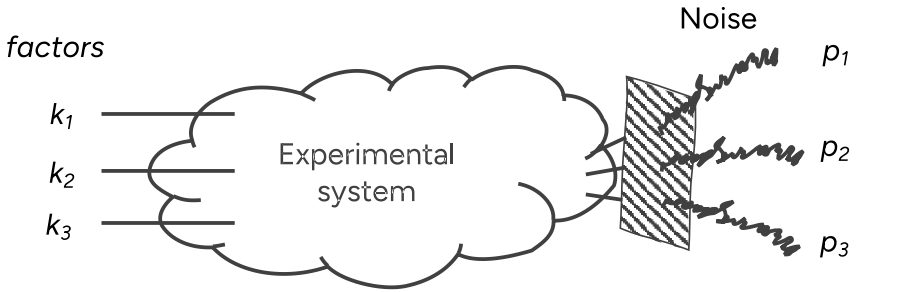
Multiple linear regression



$$\widehat{child.height} = \beta_0 + \beta_1 \cdot father.height + \beta_2 \cdot mother.height + \beta_3 \cdot grandparent.height + \cdots + \beta_j \cdot x_j$$

$H_0: \beta_1 = \beta_2 = \cdots = \beta_j = 0$

$H_1: \exists\, j \mid \beta_j \neq 0$



n entities = $k_3 \times p_3$

some model function of $y \leftarrow f(x)$

$y_1 \sim \beta_{1,1}\, x_1 + \mathcal{E}$

$y_2 \sim \beta_{2,1}\, x_1 + \beta_{2,3}\, x_3 + \mathcal{E}$

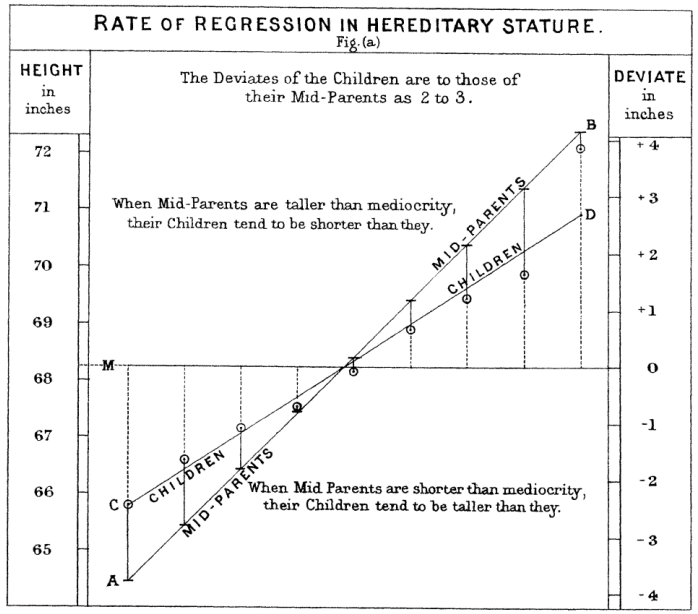
β coefficient

What is the impact of two or more independent variables, or their combined effect, on a dependent variable?

- Hypothesis testing.
- Explore and estimate causes.
- Predict effect.

Harrell 2015

Assumptions of linear regression



$$\widehat{child.height} = \beta_0 + \beta_1 \cdot parent.height$$

I. **Validity** Analysed data maps to the research question.
II. **Additivity and linearity** the relationship between x and y is linear.
III. **Indepedence of errors** errors from the prediction line are independent.
IV. **Homoscedasticity** equal variance of errors.
V. **Normality** Residual errors are normally distributed. Least important, does not substantially affect the estimation of $X \cdot \beta$.

Gelman & Hill 2007

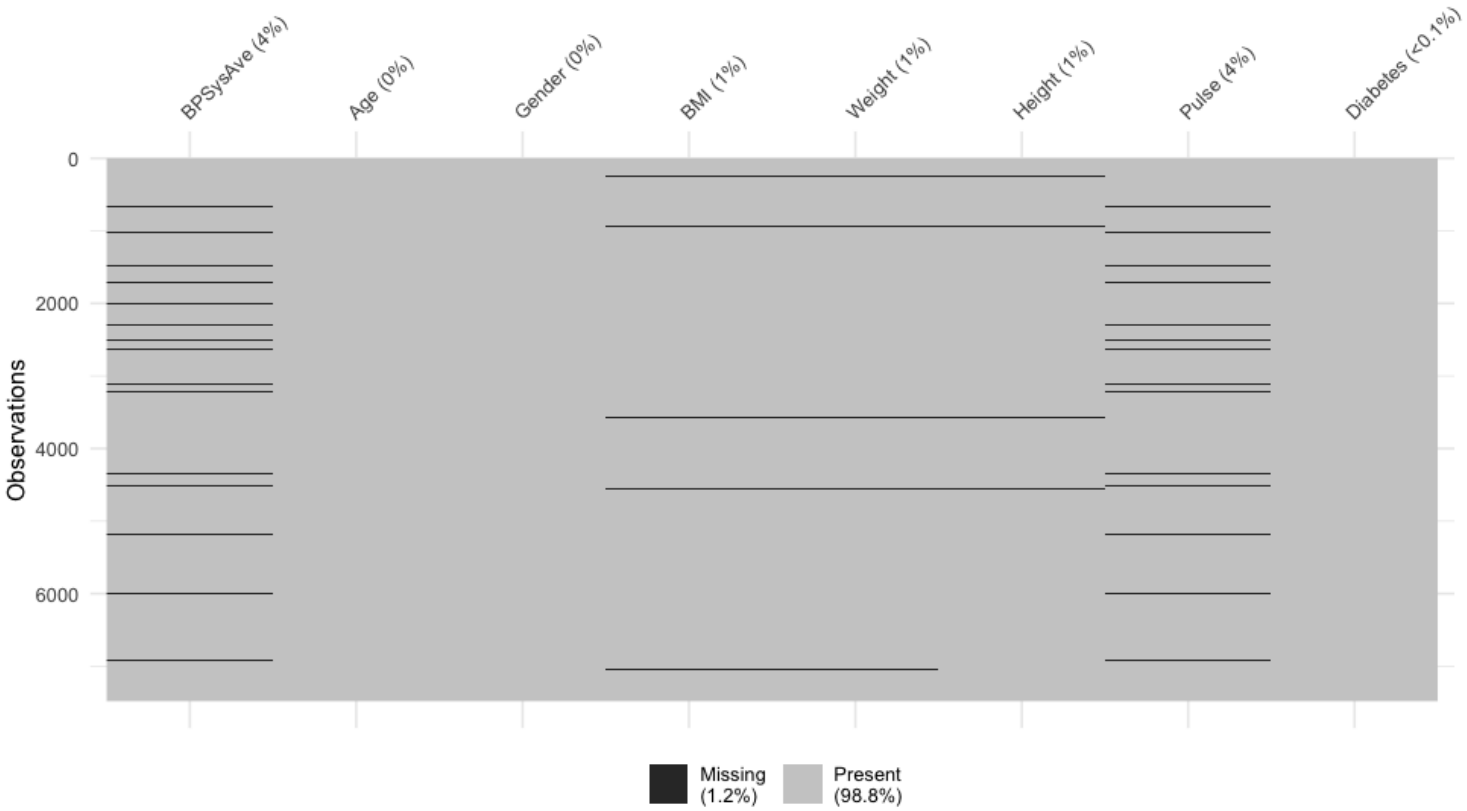
NHANES

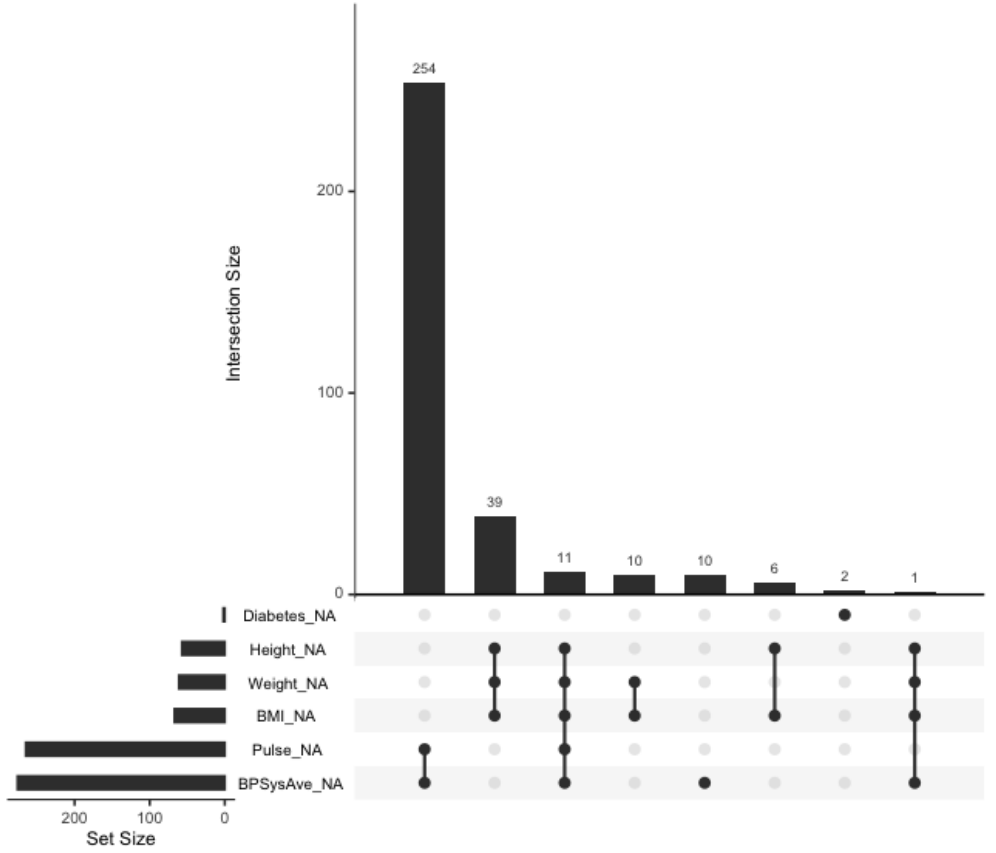
Research question In adults, 18 years and older,
What are the predictors of high systolic blood pressure?

- BPSysAve, Average systolic blood pressure.
- Age, years.
- Gender, female or male.
- BMI, kg/m^2.
- Weight, kg.
- Height, cm.
- Pulse, bpm.
- Diabetes, yes or no.

10 – 15 / m independent variables (Harrell 2015)
n observations 10,000

Missingness

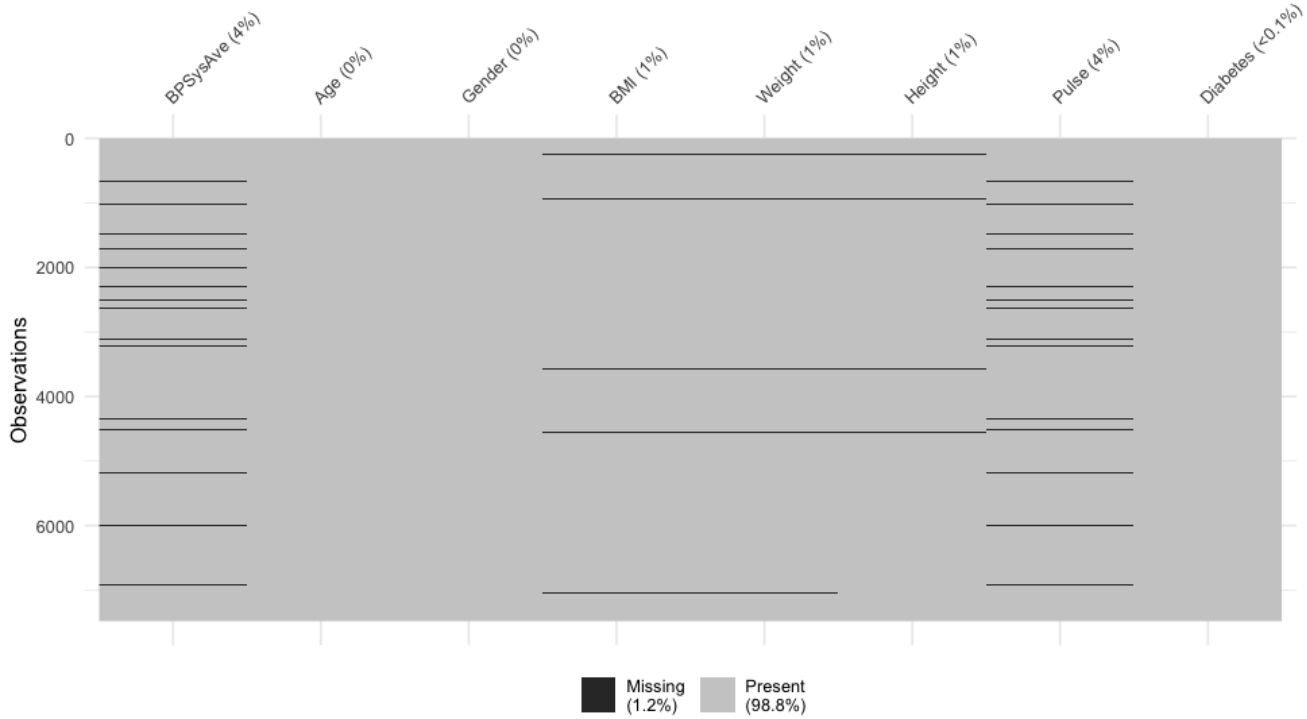




Tierney 2025 – naniar vignette - https://naniar.njtierney.com/articles/naniar.html

Missingness

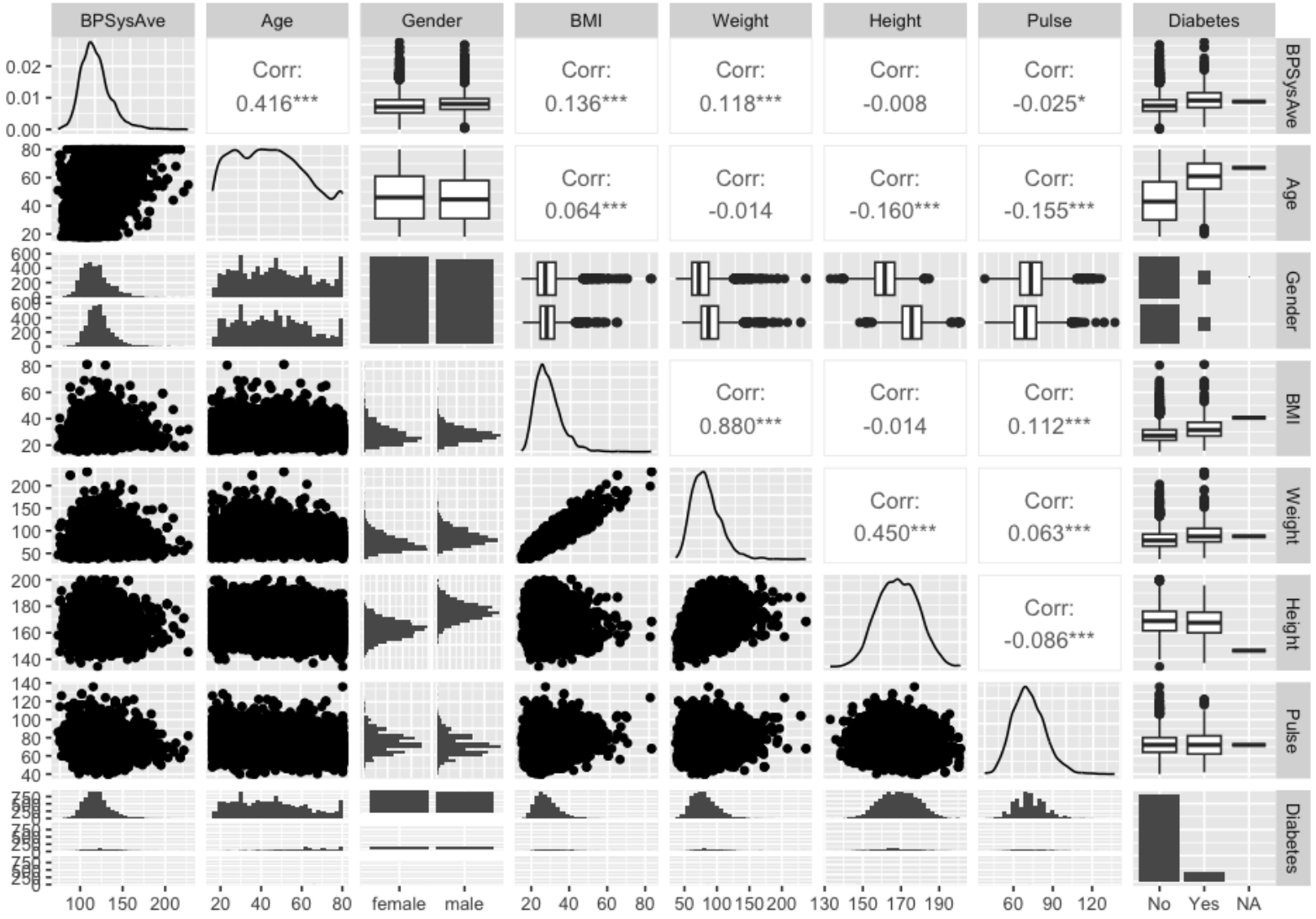
- **Missing completely at random (MCAR)** Not explained by other variables or missing data. No bias introduced.
- **Missing at random (MAR).** Explained by observed data. Allows prediction of missing data (MI).
- **Missing not at random (MNAR).** Depends on missing data.



- Complete case analysis through listwise deletion. Suitable when: missingess < 5% or > 40%, only dependent variable is missing data, certain that data is MCAR.
- Single imputation (*e.g.*, mean/median).
- Multiple imputation (regression methods).

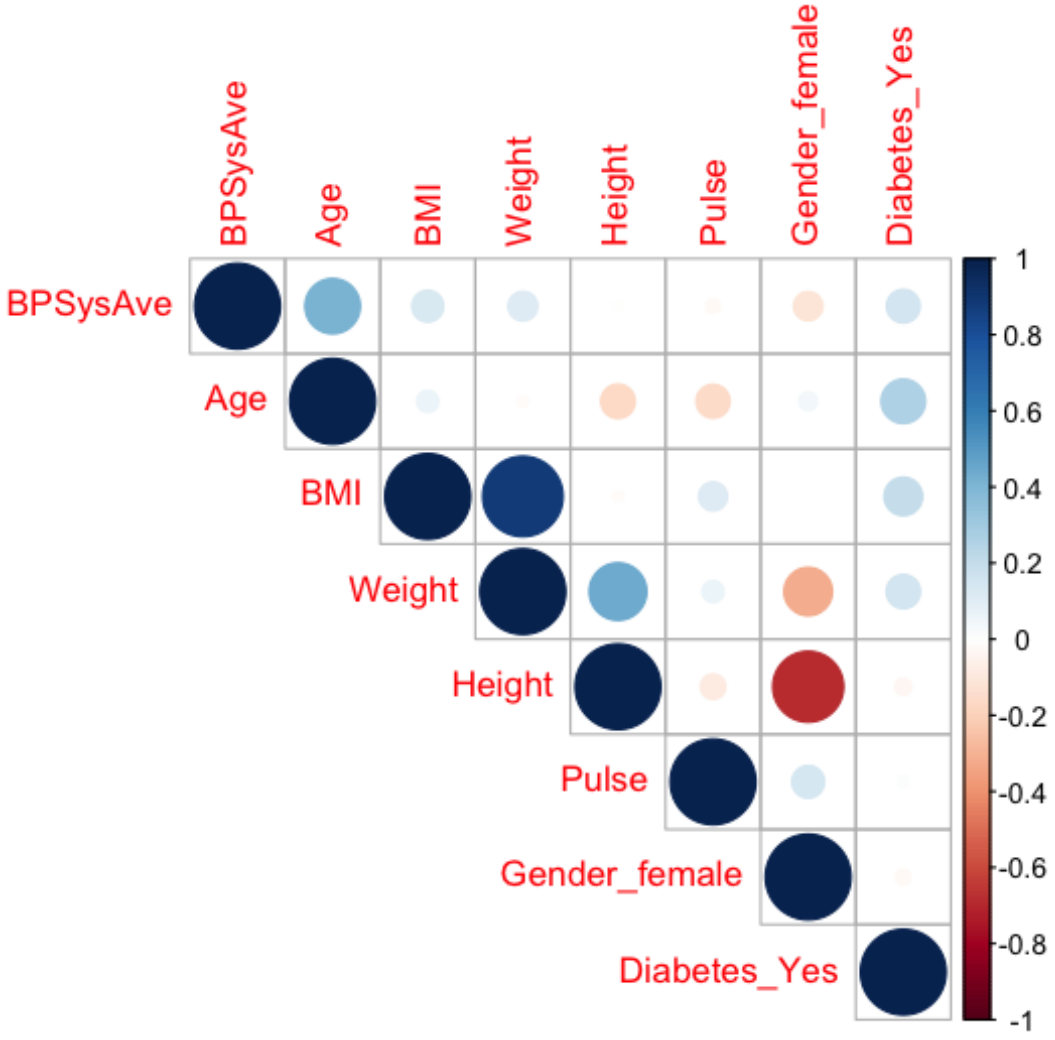
Jakobsen *et al*. 2017

Additivity & linearity



Collinearity & Identifiability

- Collinearity is when at least one predictor can be predicted well from other predictors.
- Difficult identifying the variable that is responsible for an effect.
- Inflated standard errors, reduced power. Model misspecification.



Gelman & Hill 2007; Harrell 2015

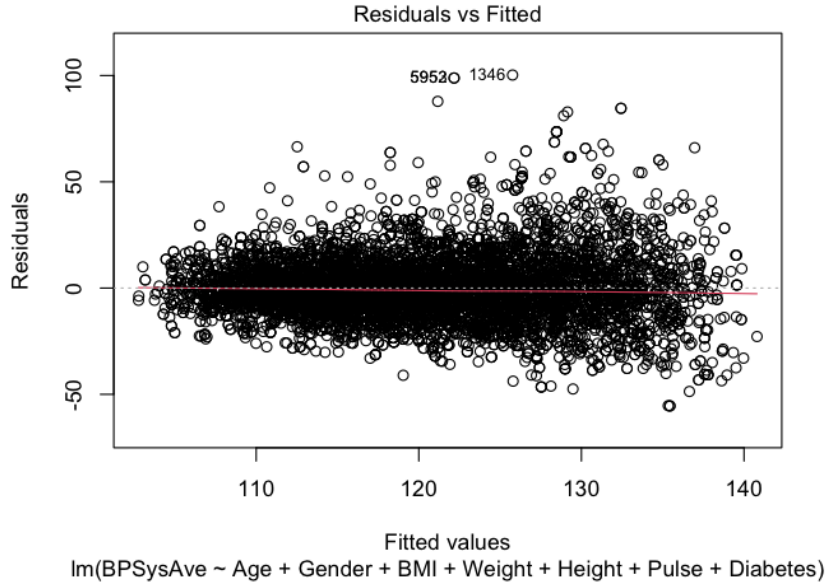
Model 1 (M_1): Variance inflation factor (VIF)

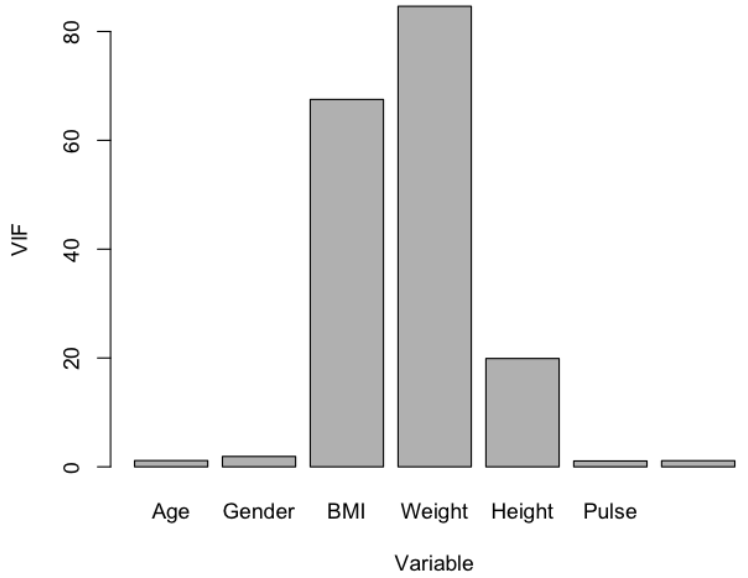
```
lm(formula = BPSysAve ~ Age + Gender + BMI + Weight +
Height + Pulse + Diabetes, data = bp_nhanes)
            coef.est coef.se
(Intercept) 106.31    13.44
Age           0.40     0.01
Gendermale    6.21     0.49
BMI           0.20     0.22
Weight        0.02     0.08
Height       -0.12     0.08
Pulse         0.07     0.02
DiabetesYes   0.82     0.63
---
n = 7148, k = 8
residual sd = 15.15, R-Squared = 0.21
```

$$VIF = \frac{1}{1 - R_i^2}$$

R_i^2 is the squared multiple correlation coefficient between column i and the remaining columns of the design matrix.
VIF over 10, consider data reduction.
`In R library(car), vif(model)`

- Reducing dataset by dropping variables.
- Creating summary scores (sum, average, regularisation, principal component).





Harrell 2015

Model 1 (M_1): Variance inflation factor (VIF)

```
lm(formula = BPSysAve ~ Age + Gender + BMI + Weight +
Height + Pulse + Diabetes, data = bp_nhanes)
             coef.est coef.se
(Intercept) 106.31    13.44
Age           0.40     0.01
Gendermale    6.21     0.49
BMI           0.20     0.22
Weight        0.02     0.08
Height       -0.12     0.08
Pulse         0.07     0.02
DiabetesYes   0.82     0.63
---
n = 7148, k = 8
residual sd = 15.15, R-Squared = 0.21
```

$$VIF = \frac{1}{1 - R_i^2}$$

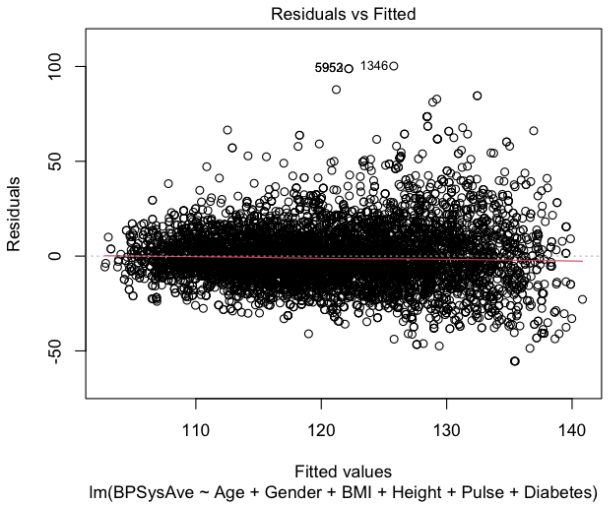
R_i^2 is the squared multiple correlation coefficient between column i and the remaining columns of the design matrix.
VIF over 10, consider data reduction.
`In R library(car), vif(model)`

- Reducing dataset by dropping variables.
- Creating summary scores.

Harrell 2015

M_2: Reduced dataset

```
lm(formula = BPSysAve ~ Age + Gender + BMI + Height + Pulse +
    Diabetes, data = bp_nhanes)
             coef.est coef.se
(Intercept) 103.38     4.35
Age           0.40     0.01
Gendermale    6.21     0.49
BMI           0.25     0.03
Height       -0.10     0.02
Pulse         0.07     0.02
DiabetesYes   0.81     0.63
---
n = 7148, k = 7
residual sd = 15.15, R-Squared = 0.21
```



M_3: Interactions

```
lm(formula = BPSysAve ~ Age + Gender +
Age * Gender + BMI + Height +
    Pulse + Diabetes, data = bp_nhanes)
               coef.est coef.se
(Intercept)     95.29     4.35
Age              0.53     0.01
Gendermale      18.07     1.06
BMI              0.25     0.03
Height          -0.09     0.02
Pulse            0.08     0.02
DiabetesYes      1.00     0.63
Age:Gendermale -0.26      0.02
```

- Two-way interactions alone are often too many to allow testing of all at a reasonable power.
- Large main effects may also exhibit interaction effects.
- A priori identified interection effects.

- *Interactions between treatment and severity of disease.*
- *Interactions between age and risk factors.*
- *Interactions between age and disease.*
- *Interactions between measurement and state of individual.*
- *Interactions between time and treatment (learning effects).*

Gelman & Hill 2007; Harrell 2015

Centering and/or standardising data

```
Age <- Age – mean(age)
```

Centering Linear transformations, such as the above, do not affect residual errors or R^2. They can simplify the interpretation of regressions, differences from the mean.

```
Age <- (Age – mean(age))/(2*sd(Age))
```

Standardising Scaling of predictors allows comparisons across coefficients on approximately the same scale.

Building regression models for hypothesis testing

- Work with the full model, including insignificant variables.
- Carefully consider what interaction effects to consider and how these should be tested (*e.g.*, blocks of interactions or main effect and interaction effect).
- Consider the effect of imputing missing values.
- Consider the degrees of freedom to test the predictor of interest.
- Should not shrink parameter estimates of interest for hypothesis testing.
- Model validation is usually not necessary unless you wish to assess the degree of overfitting.

Harrell 2015

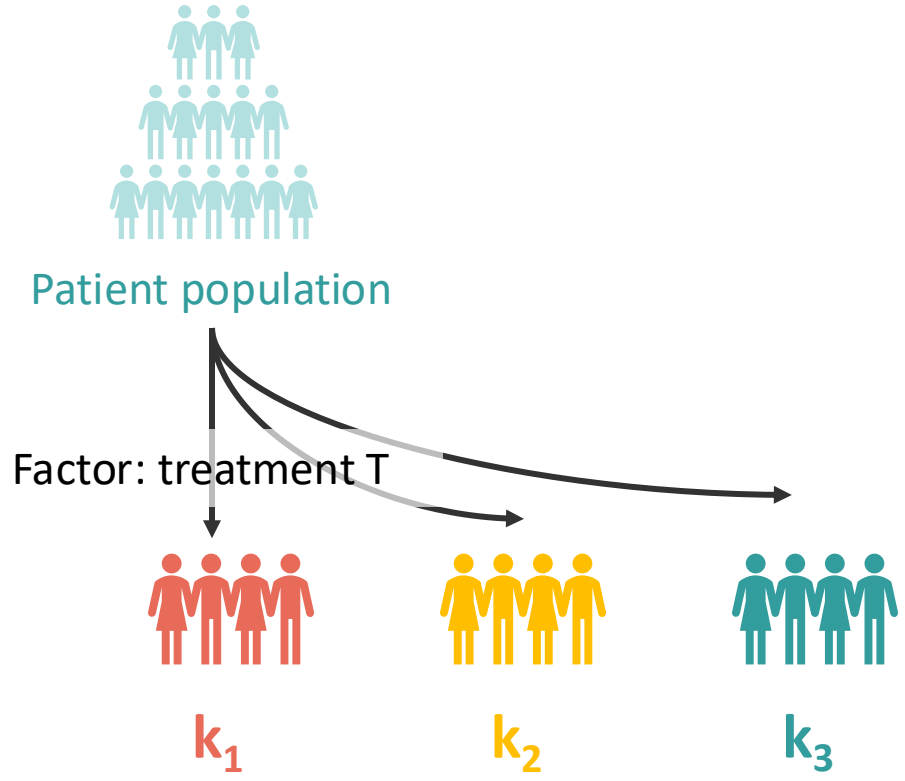
Building regression models for prediction

- Include all input variables that might affect the outcome.
- Several predictors can be combined into a sum or average score.
- For inputs with large effects, consider their interactions.
- On the exclusion of variables:
  - If predictor is not statistically significant and has the expected sign, leave it in the model.
  - If a predictor is not statistically significant and does not have the expected sign, consider removing it.
  - If a predictor is statistically significant and does not have the expected sign, think hard and consider collecting more data.
  - If a predictor is statistically significant and has the expected sign, keep it in.

Gelman & Hill 2007

L2-8 ANOVA Analysis of Variance

Factorial Experiment Design



k levels: treatments

1-way ANOVA

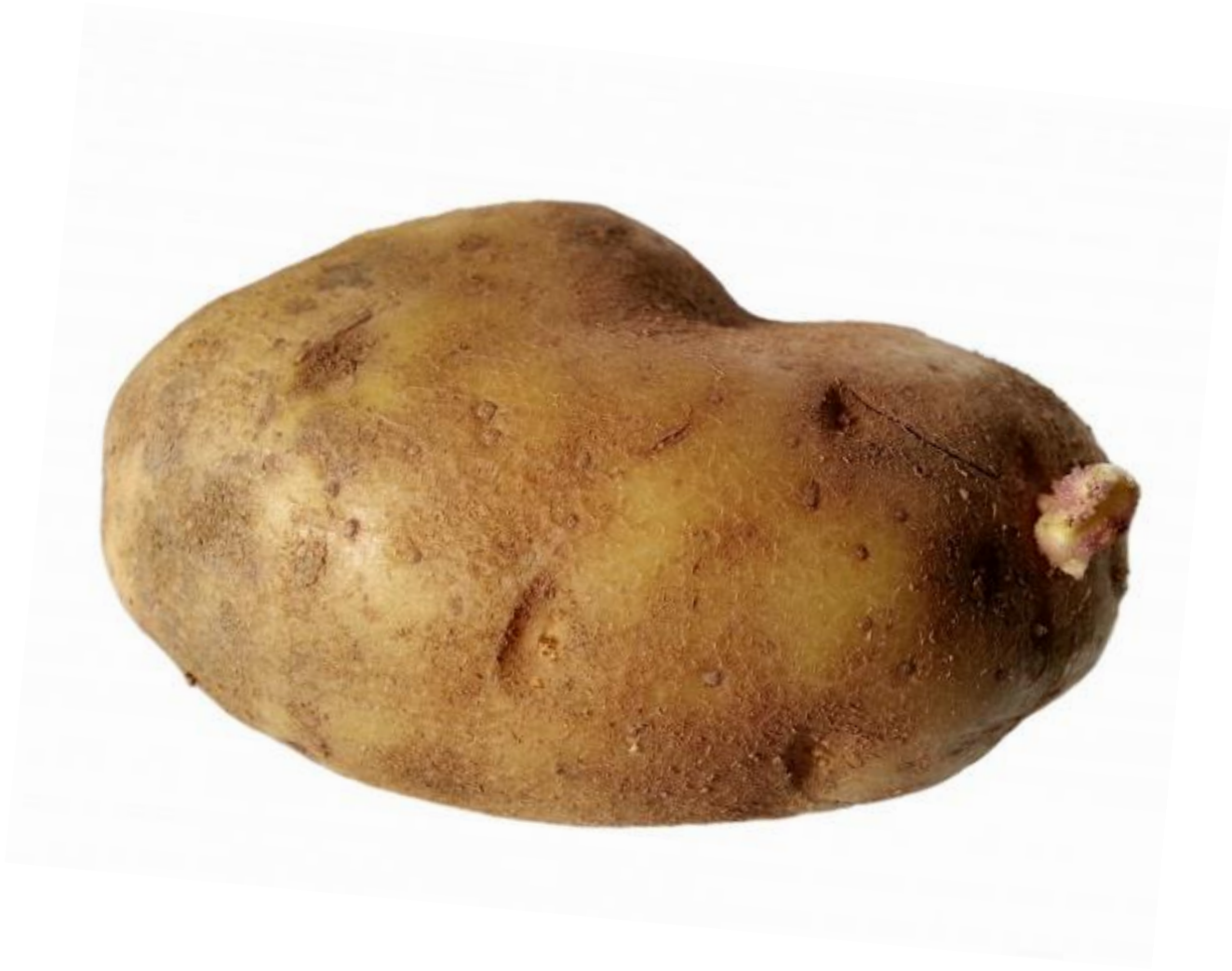
Comparing the average effects elicited by *k* different levels of a given factor where k is equal to two or more.

H_0: $\mu_1 = \mu_2 = \cdots = \mu_k$

H_1: $not\ all\ \mu_j s\ are\ equal$

Treatment levels

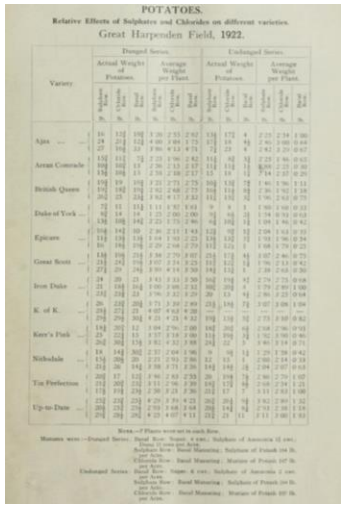
	T_1	T_2	T_k
	Y_{11}	Y_{12}	Y_{1k}
	⋮	⋮	⋮
	$Y_{n_1}1$	$Y_{n_2}2$	$Y_{n_k}k$
Sample sizes	n_1	n_2	n_k
Sample totals	$T_{.1}$	$T_{.2}$	$T_{.k}$
Sample means	$\bar{Y}_{.1}$	$\bar{Y}_{.2}$	$\bar{Y}_{.k}$
True means	μ_1	μ_2	μ_k



ANOVA: Analysis of Variance



THE NEW ROTHAMSTED LABORATORIES, ERECTED 1914-1916



POTATOES.

Relative Effects of Sulphates and Chlorides on different varieties.

Great Harpenden Field, 1922.

Variety	Dunged Series. Actual Weight of Potatoes.	Dunged Series. Average Weight per Plant.	Unchanged Series. Actual Weight of Potatoes.	Unchanged Series. Average Weight per Plant.
	lbs.	lbs.	lbs.	lbs.
Ajax	15 132 109	1·36 2·75 2·02	132 172 4	2·75 3·54 1·00
Arran Comrade	24 24 21	1·60 1·60 1·40	16 16 16	1·06 1·06 1·06
British Queen	27 104 33	1·80 6·93 2·20	23 8 2	1·53 0·53 0·13
Duke of York	132 132 112	1·94 1·94 1·65	132 48 72	1·94 0·71 1·06
Epicure	15 15 13	1·00 1·00 0·87	15 15 15	1·00 1·00 1·00
Great Scott	132 132 109	1·94 1·94 1·60	132 132 132	1·94 1·94 1·94
Iron Duke	132 132 132	1·94 1·94 1·94	132 132 132	1·94 1·94 1·94
K. of K.	24 24 24	1·60 1·60 1·60	24 24 24	1·60 1·60 1·60
Kerr's Pink	132 132 132	1·94 1·94 1·94	132 132 132	1·94 1·94 1·94
Nithsdale	132 132 132	1·94 1·94 1·94	132 132 132	1·94 1·94 1·94
Tinwald Perfection	132 132 132	1·94 1·94 1·94	132 132 132	1·94 1·94 1·94
Up-to-Date	132 132 132	1·94 1·94 1·94	132 132 132	1·94 1·94 1·94

Means were—Dunged Series. Basal Row: Sugars 4 cwt., Sulphate of Ammonia 12 cwt., Dunged 12 tons per Acre. Sulphate Row: Basal Manuring, Sulphate of Potash 168 lb. per Acre. Chloride Row: Basal Manuring, Muriate of Potash 167 lb. per Acre. Unchanged Series. Basal Row: Sugars 4 cwt., Sulphate of Ammonia 4 cwt. per Acre. Sulphate Row: Basal Manuring, Sulphate of Potash 168 lb. per Acre. Chloride Row: Basal Manuring, Muriate of Potash 167 lb. per Acre.

AJAX	K OF K	NITHSDALE	GREAT SCOTT	DUKE OF YORK	S
GREAT SCOTT	DUKE OF YORK	ARRAN COMRADE	IRON DUKE	EPICURE	C
IRON DUKE	EPICURE	AJAX	K OF K	NITHSDALE	B
K OF K	NITHSDALE	GREAT SCOTT	DUKE OF YORK	ARRAN COMRADE	S
	UP TO DATE	KERR'S PINK	UP TO DATE	BRITISH QUEEN	C
	BRITISH QUEEN	TINWALD PERFECTION	EPICURE	KERR'S PINK	B
	KERRS PINK	UP TO DATE	IRON DUKE	AJAX	S
	TINWALD PERFECTION	ARRAN COMRADE	BRITISH QUEEN	TINWALD PERFECTION	C
					B

S = SULPHATE ROW C = CHLORIDE ROW
B = BASAL ROW

Diagram 1. Plan of experiment. Farmyard manure series.

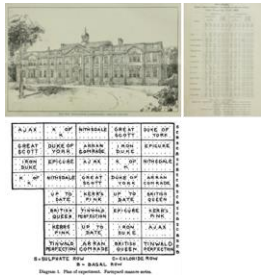
When the mean square deviation is significantly larger than the standard deviation of the differences we may conclude that the corresponding effect is not due to chance.



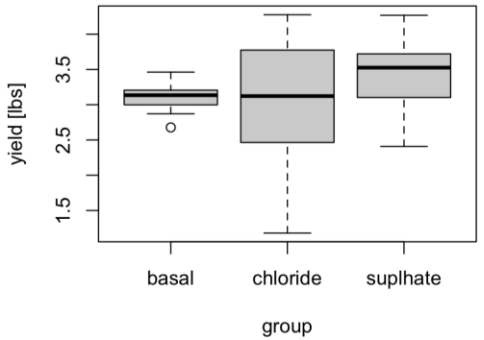
Ronald Fisher

Fisher & MacKenzie 1923; Rothamsted Experimental Station Harpenden 1923

ANOVA: Analysis of Variance



	Basal	Chloride	Sulphate
Sample sizes (n)	*21*	*21*	*21*
Sample totals ($T_{\cdot}$)	*64.97*	*62.70*	*72.10*
Sample means ($\overline{Y_{\cdot j}}$)	*3.09*	*2.98*	*3.43*
True means (μ)	μ_{k1}	μ_{k2}	μ_{k3}



$$\bar{Y}_{\cdot\cdot} = \frac{1}{n}\sum_{j=1}^{k} T_{\cdot j}$$

$$SSTR = \sum_{j=1}^{k} n_j \left(\bar{Y}_{\cdot j} - \bar{Y}_{\cdot\cdot}\right)^2$$

$$SSE = \sum_{j=1}^{k}\sum_{i=1}^{n_j} \left(Y_{ij} - \bar{Y}_{\cdot j}\right)^2$$

ANOVA table

Source	*df*	*SS*	*MS*	*F*	*P*
Treatment	$k-1$	$SSTR$	$MSTR = SSTR\ /\ (k-1)$	$MSTR/MSE$	$P(F_{(k-1,n-k)} \geq observed\ F)$
Error	$n-k$	SSE	$MSE = SSE\ /\ (n-k)$		
Total	$n-1$	$SSTOT = SSTR + SSE$			

SSE: error sum of squares; SSTR: treatment sum of squares;
SSTOT: total variation; $\bar{Y}$: overall sample mean; MSTR: mean square treatment;
MSE: mean squared error

Fisher & MacKenzie 1923; Rothamsted Experimental Station Harpenden 1923; Larsen & Marx 2007

ANOVA: Analysis of Variance

ANOVA table

Source	*df*	*SS*	*MS*	*F*	*P*
Treatment	$2 = k - 1$	$SSTR = 2.30$	$MSTR = \frac{SSTR}{(k-1)} = 1.15$	$\frac{MSTR}{MSE} = 3.20$	$P(F_{(k-1,n-k)} \geq observed\ F) = 0.048$
Error	$60 = n - k$	$SSE = 21.57$	$MSE = \frac{SSE}{(n-k)} = 0.36$		
Total	$62 = n - 1$	$SSTOT = SSTR + SSE = 23.87$			

```
m1_aov <- aov(yield_lbs ~ group, greatscott)
summary(m1_aov) OR
lm1_aov <- lm(yield_lbs ~ group, greatscott)
anova(m1_greatscott)

Analysis of Variance Table

Response: yield_lbs
          Df  Sum Sq Mean Sq F value  Pr(>F)
group      2  2.3015 1.15075  3.2014 0.04775 *
Residuals 60 21.5675 0.35946
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

ANOVA: Analysis of Variance

ANOVA table

Source	*df*	*SS*	*MS*	*F*	*P*
Treatment	$2 = k - 1$	$SSTR = 2.30$	$MSTR = \frac{SSTR}{(k-1)} = 1.15$	$\frac{MSTR}{MSE} = 3.20$	$P(F_{(k-1,n-k)} \geq observed\ F) = 0.048$
Error	$60 = n - k$	$SSE = 21.57$	$MSE = \frac{SSE}{(n-k)} = 0.36$		
Total	$62 = n - 1$	$SSTOT = SSTR + SSE = 23.87$			

```
m1_aov <- aov(yield_lbs ~ group, greatscott)
summary(m1_aov) OR
lm1_aov <- lm(yield_lbs ~ group, greatscott)
anova(m1_greatscott)

Analysis of Variance Table

Response: yield_lbs
          Df  Sum Sq Mean Sq F value  Pr(>F)
group      2  2.3015 1.15075  3.2014 0.04775 *
Residuals 60 21.5675 0.35946
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
OR summary(m1_greatscott)

Call:
lm(formula = yield_lbs ~ group, data = greatscott)

Residuals:
     Min       1Q   Median       3Q      Max
-1.80477 -0.24978  0.06154  0.22876  1.29119

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     3.0938     0.1308  23.647   <2e-16 ***
groupchloride  -0.1090     0.1850  -0.589   0.5579
groupsuplhate   0.3398     0.1850   1.836   0.0712 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.5995 on 60 degrees of freedom
Multiple R-squared:  0.09642,    Adjusted R-squared:  0.0663
F-statistic: 3.201 on 2 and 60 DF,  p-value: 0.04775
```

Post-hoc analysis: Tukey HSD

Tukey HSD (honest significant difference) is one of several tests for the purpose of pairwise comparisons of means. The test maintains alpha of 0.05 across multiple comparisons.
Post hoc tests should only be applied after a significant overall effect has been found (omnibus test).

```
TukeyHSD(m1_aov)

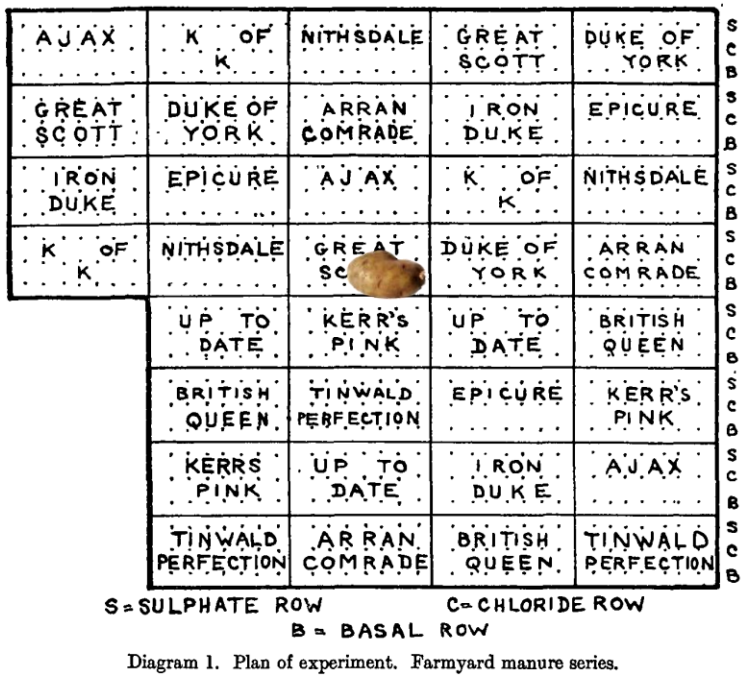
 Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = yield_lbs ~ group, data = greatscott)

$group
                       diff          lwr       upr     p adj
chloride-basal    -0.1090313 -0.553685157 0.3356225 0.8263814
suplhate-basal     0.3397909 -0.104862958 0.7844447 0.1665540
suplhate-chloride  0.4488222  0.004168382 0.8934760 0.0474110
```



ANOVA: Analysis of Variance



AJAX	K OF K	NITHSDALE	GREAT SCOTT	DUKE OF YORK	S C B
GREAT SCOTT	DUKE OF YORK	ARRAN COMRADE	IRON DUKE	EPICURE	S C B
IRON DUKE	EPICURE	AJAX	K OF K	NITHSDALE	S C B
K OF K	NITHSDALE	GREAT SC	DUKE OF YORK	ARRAN COMRADE	S C B
	UP TO DATE	KERR'S PINK	UP TO DATE	BRITISH QUEEN	S C B
	BRITISH QUEEN	TINWALD PERFECTION	EPICURE	KERR'S PINK	S C B
	KERRS PINK	UP TO DATE	IRON DUKE	AJAX	S C B
	TINWALD PERFECTION	ARRAN COMRADE	BRITISH QUEEN	TINWALD PERFECTION	S C B

S = SULPHATE ROW C = CHLORIDE ROW
B = BASAL ROW

Diagram 1. Plan of experiment. Farmyard manure series.

```
lm(yield_lbs ~ group, greatscott)
```

ANOVA: Analysis of Variance

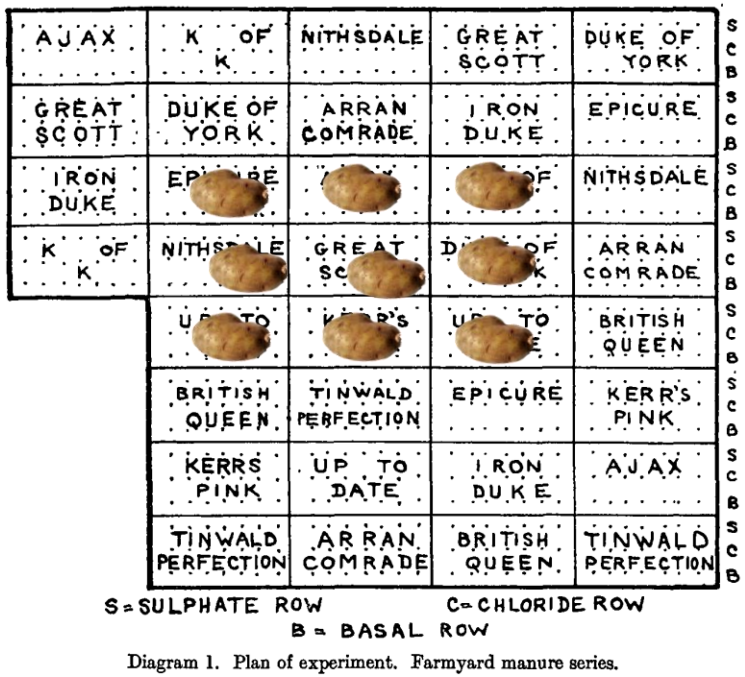


Diagram 1. Plan of experiment. Farmyard manure series.

```
lm(yield_lbs ~ group, greatscott)
```

```
lm(yield_lbs ~ manure + variety, potatoes)
```

ANOVA: Analysis of Variance

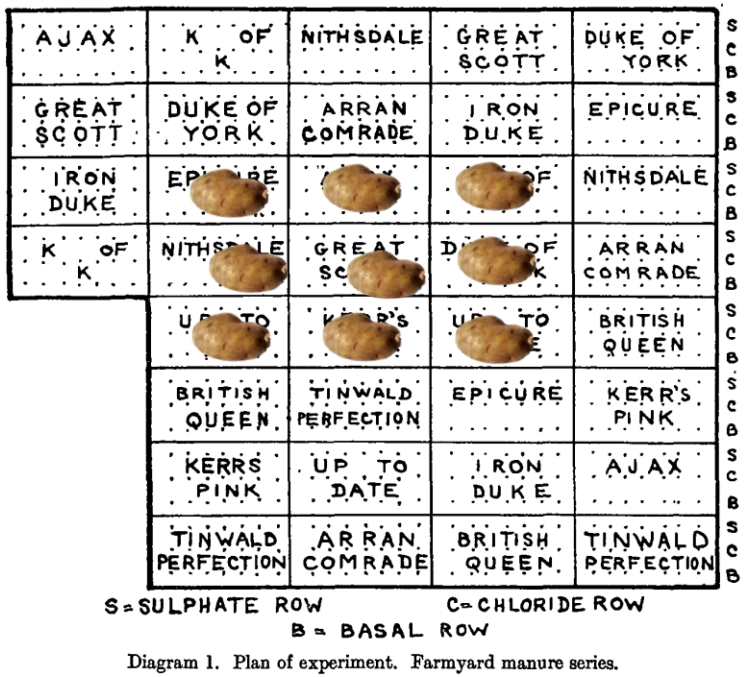
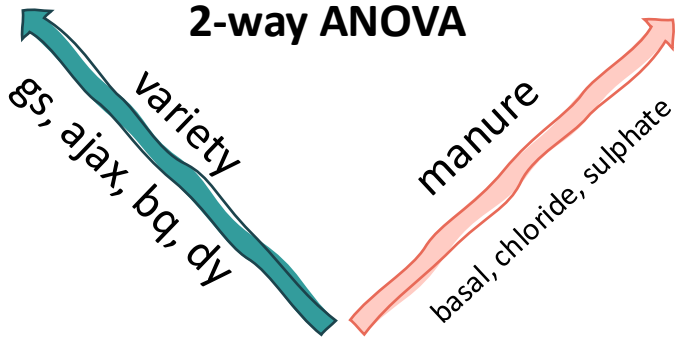


Diagram 1. Plan of experiment. Farmyard manure series.




```
m2_aov <- aov(yield_lbs ~ manure + variety, potatoes)
tukeyHSD(m2_aov)
Analysis of Variance Table

Response: yield_lbs
            Df  Sum Sq Mean Sq F value    Pr(>F)
group        2   6.123   3.062  5.2511  0.005845 **
variety      3 122.573  40.858 70.0794 < 2.2e-16 ***
Residuals  246 143.422   0.583
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Tukey multiple comparisons of means
    95% family-wise confidence level

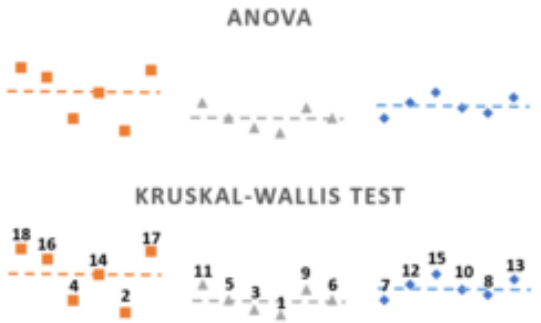
Fit: aov(formula = yield_lbs ~ group + variety, data = df_potatoes)

$group
                       diff         lwr        upr     p adj
chloride-basal    -0.2561484 -0.53396480 0.02166791 0.0776888
suplhate-basal     0.1171412 -0.16067514 0.39495757 0.5811918
suplhate-chloride  0.3732897  0.09547331 0.65110601 0.0048994

$variety
                                diff         lwr         upr     p adj
britishqueen-ajax       -0.2695082 -0.62141505  0.08239871 0.1979640
dukeofyork-ajax         -1.6706266 -2.02253346 -1.31871970 0.0000000
greatscott-ajax          0.0232889 -0.32861798  0.37519578 0.9982127
dukeofyork-britishqueen -1.4011184 -1.75302529 -1.04921153 0.0000000
greatscott-britishqueen  0.2927971 -0.05910981  0.64470395 0.1397782
greatscott-dukeofyork    1.6939155  1.34200860  2.04582236 0.0000000
```

Nonparametric alternative: Kruskal-Wallis

- The Kruskal-Wallis test is an extension of the Wilcoxon-Mann-Whitney test to handle more than two groups.
- The Kruskal-Wallis test is a so called nonparametric alternative to a one-way ANOVA relying on rank transformation, rank sum, and a test statistic similar to SSTR.
- Posthoc analysis can be done using Dunn's test, Wilcoxon, or the more powerful Conover-Iman test.




```
kruskal.test(yield_lbs ~ group, df_ajax)

        Kruskal-Wallis rank sum test

data:  yield_lbs by group
Kruskal-Wallis chi-squared = 6.9603, df = 2,
p-value = 0.0308
```

```
> library(conover.test)
> conover.test(df_ajax$yield_lbs, df_ajax$group)
  Kruskal-Wallis rank sum test

data: x and group
Kruskal-Wallis chi-squared = 6.9603, df = 2, p-value = 0.03


                           Comparison of x by group
                                (No adjustment)
Col Mean-|
Row Mean |      basal   chloride
---------+----------------------
chloride |   2.513676
         |    0.0073*
         |
suplhate |   0.281250  -2.232426
         |     0.3897    0.0147*

alpha = 0.05
Reject Ho if p <= alpha/2
```

Conover & Iman 1981